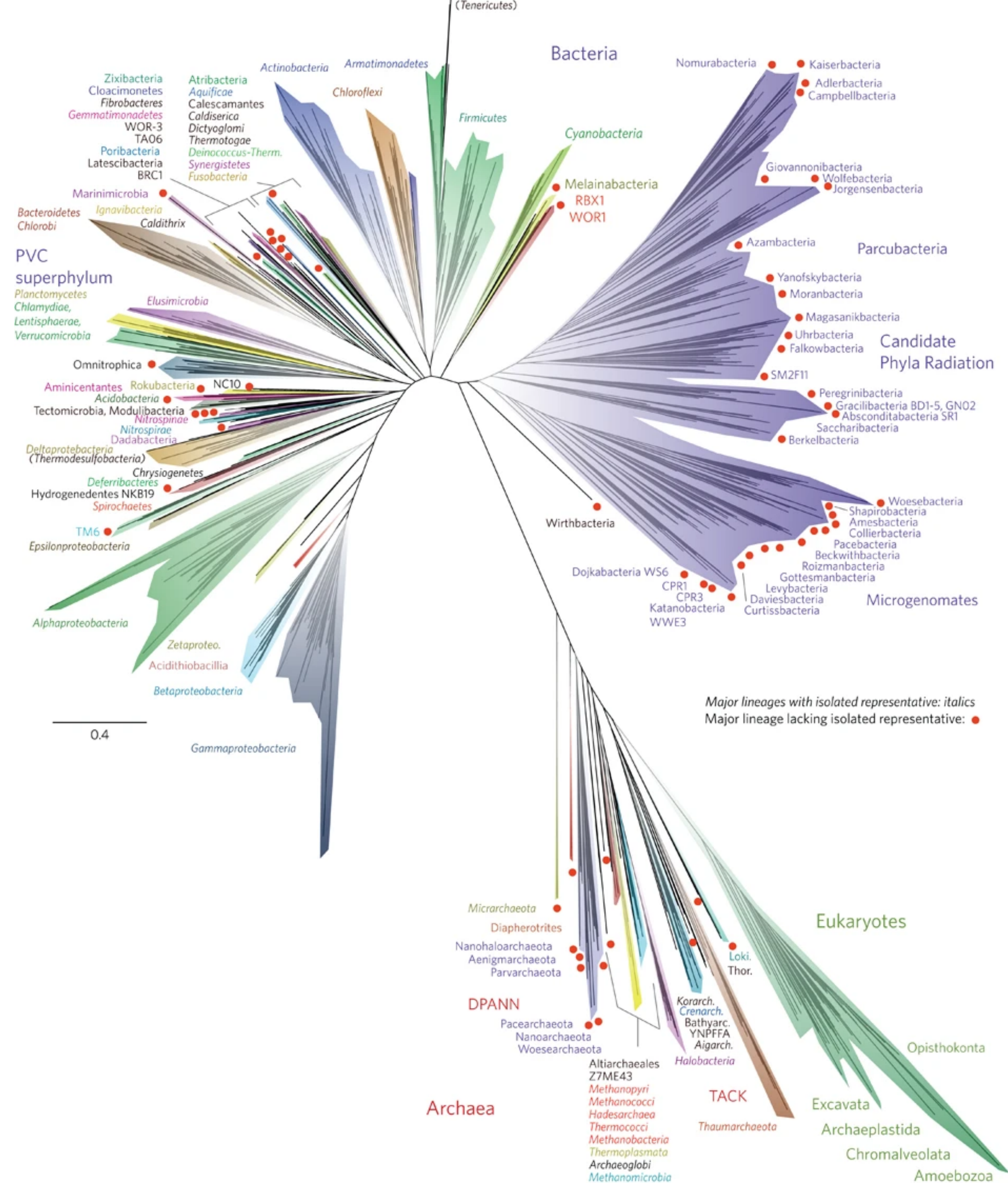


Quantitative Microbial Physiology

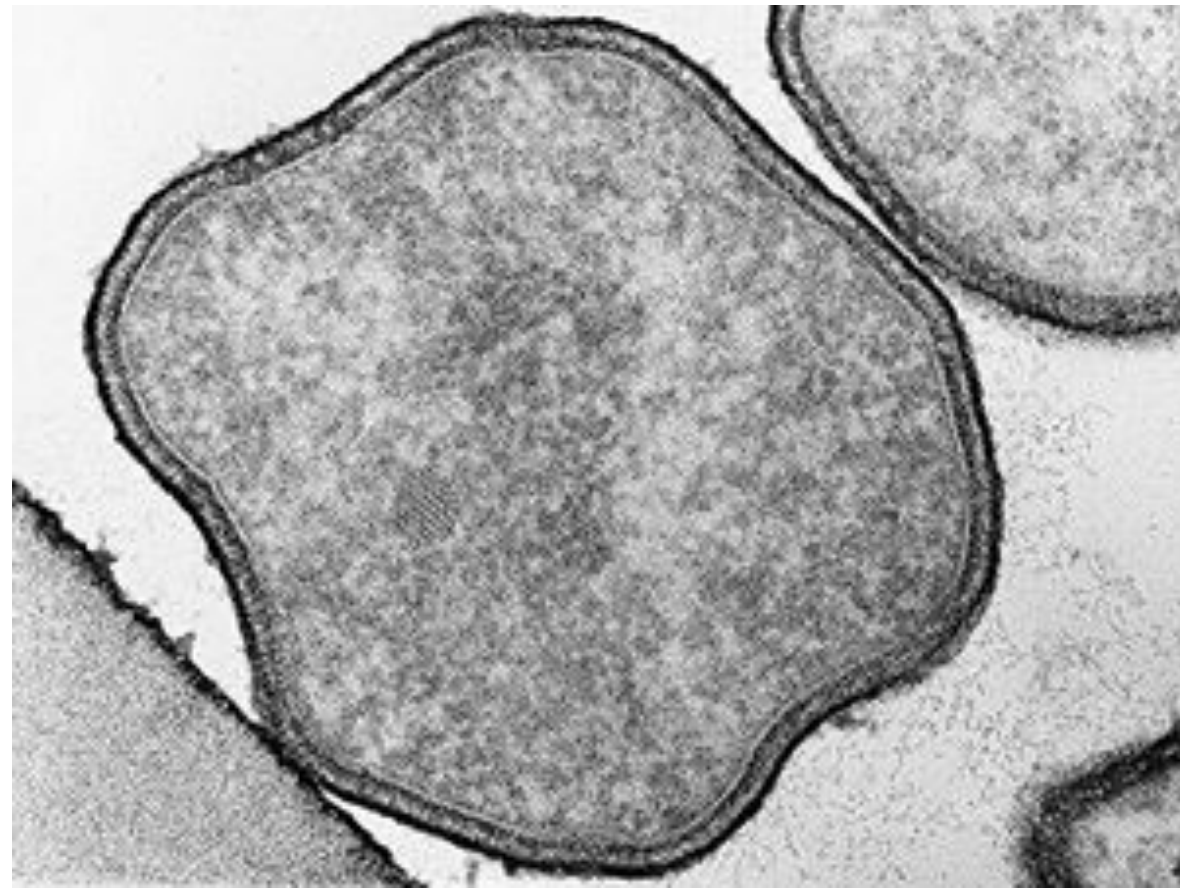
Jacopo Grilli
Lecture 1, Feb 17, 2025

Why microbes?

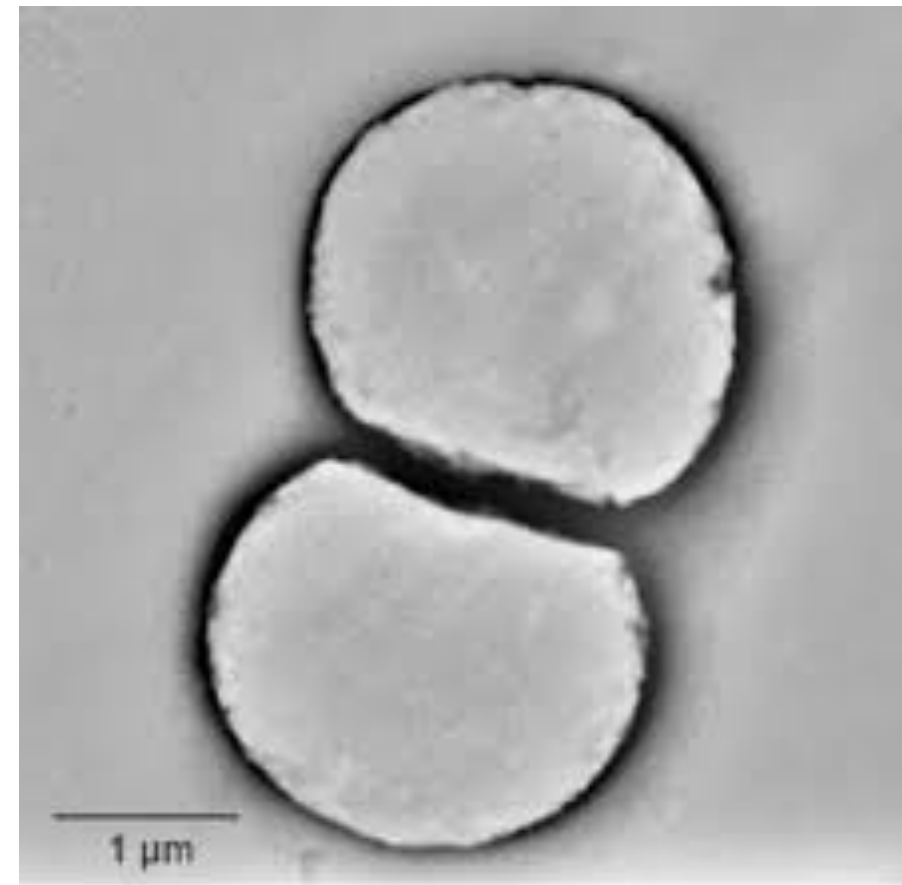
Microbes are are extremely diverse



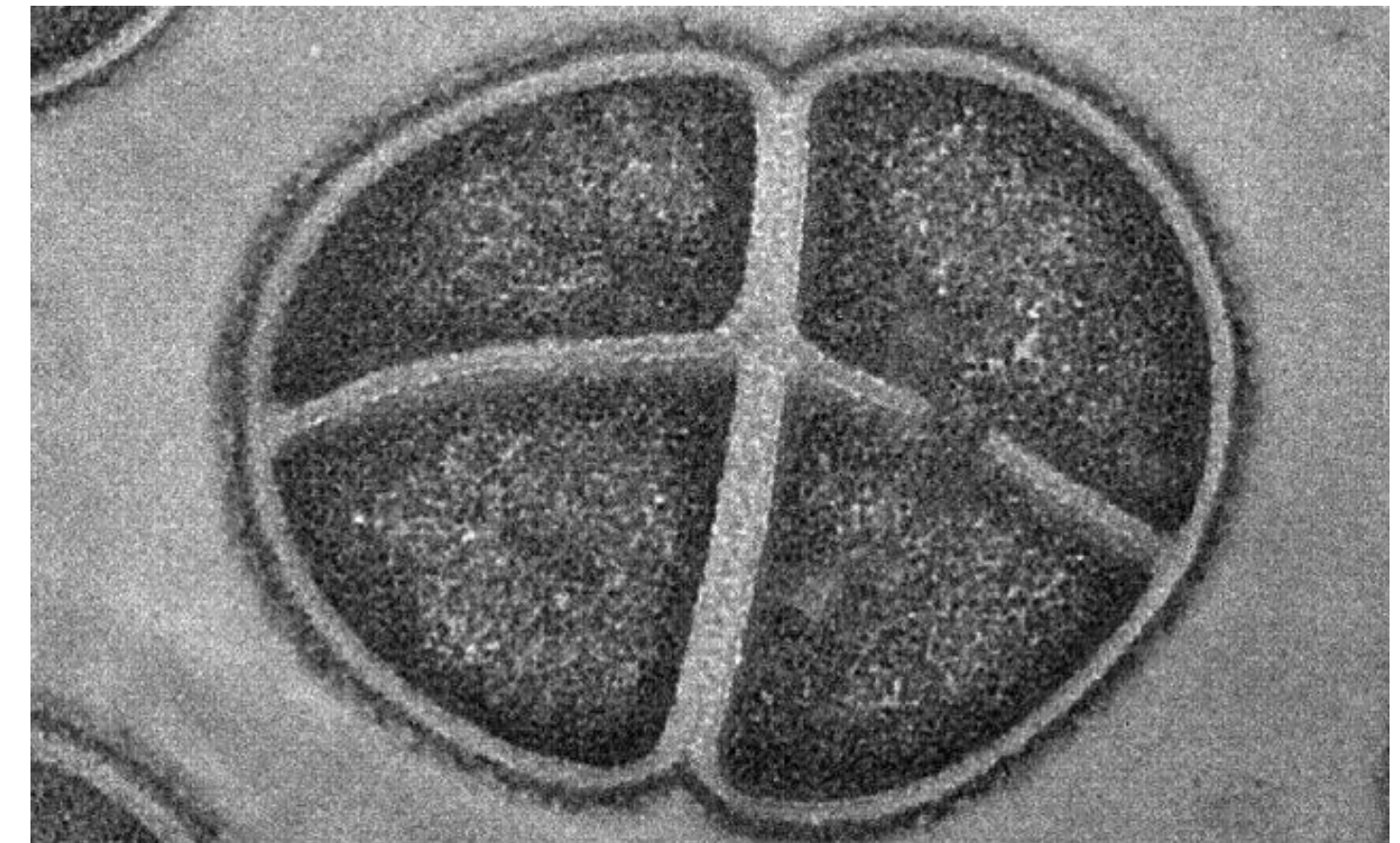
Microbes are everywhere



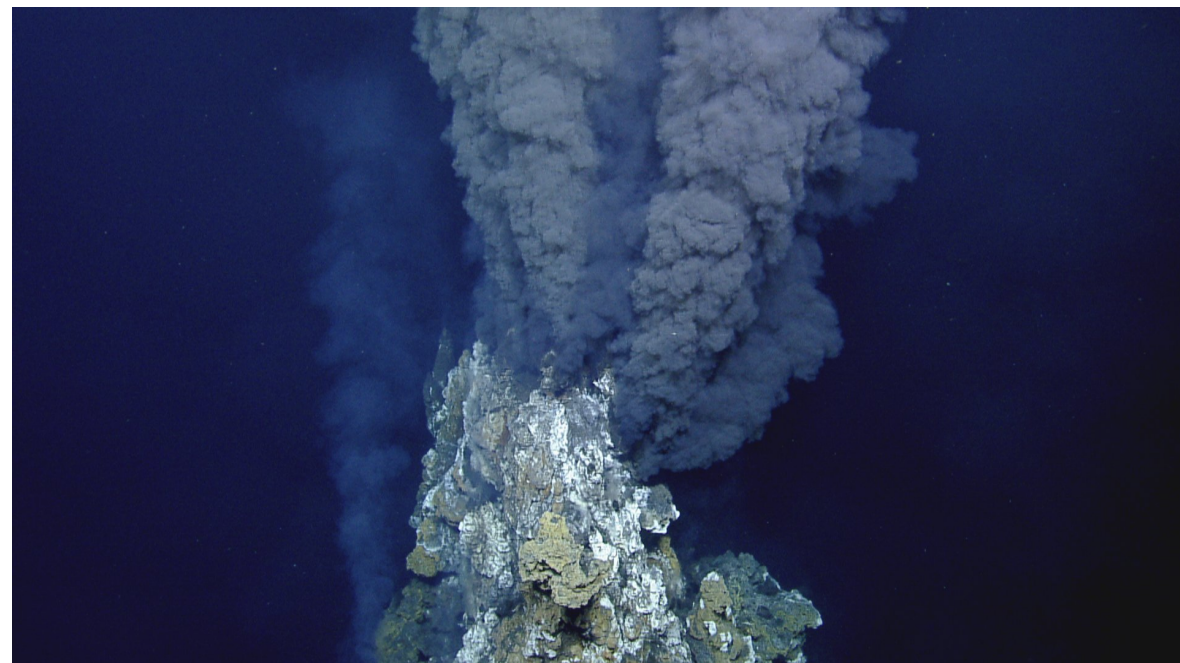
Pyrolobus fumarii
optimal growth 106°C (60 mins)
growth T in [90-113]°C
in hydrothermal vents



Picrophilus torridus
optimal growth 60°C and
pH 0.7 (6 hours)
in hot springs



Deinococcus radiodurans
survives easily 5000Gy of ionizing radiation
cold, dehydration, vacuum (3yrs in outer space)
found everywhere...

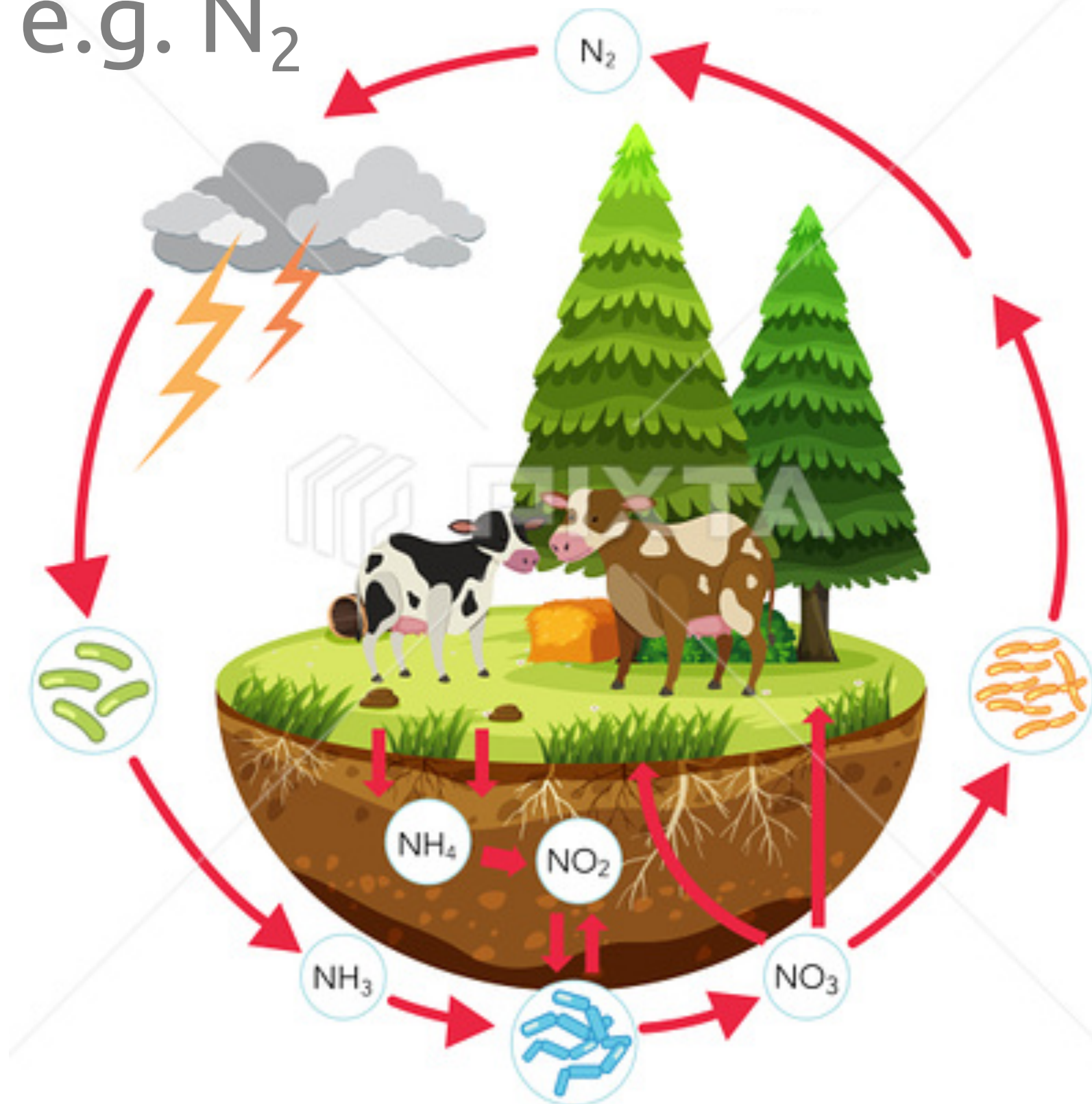


Microbes are important

“Life would not long remain possible in the absence of microbes.”*—L. Pasteur

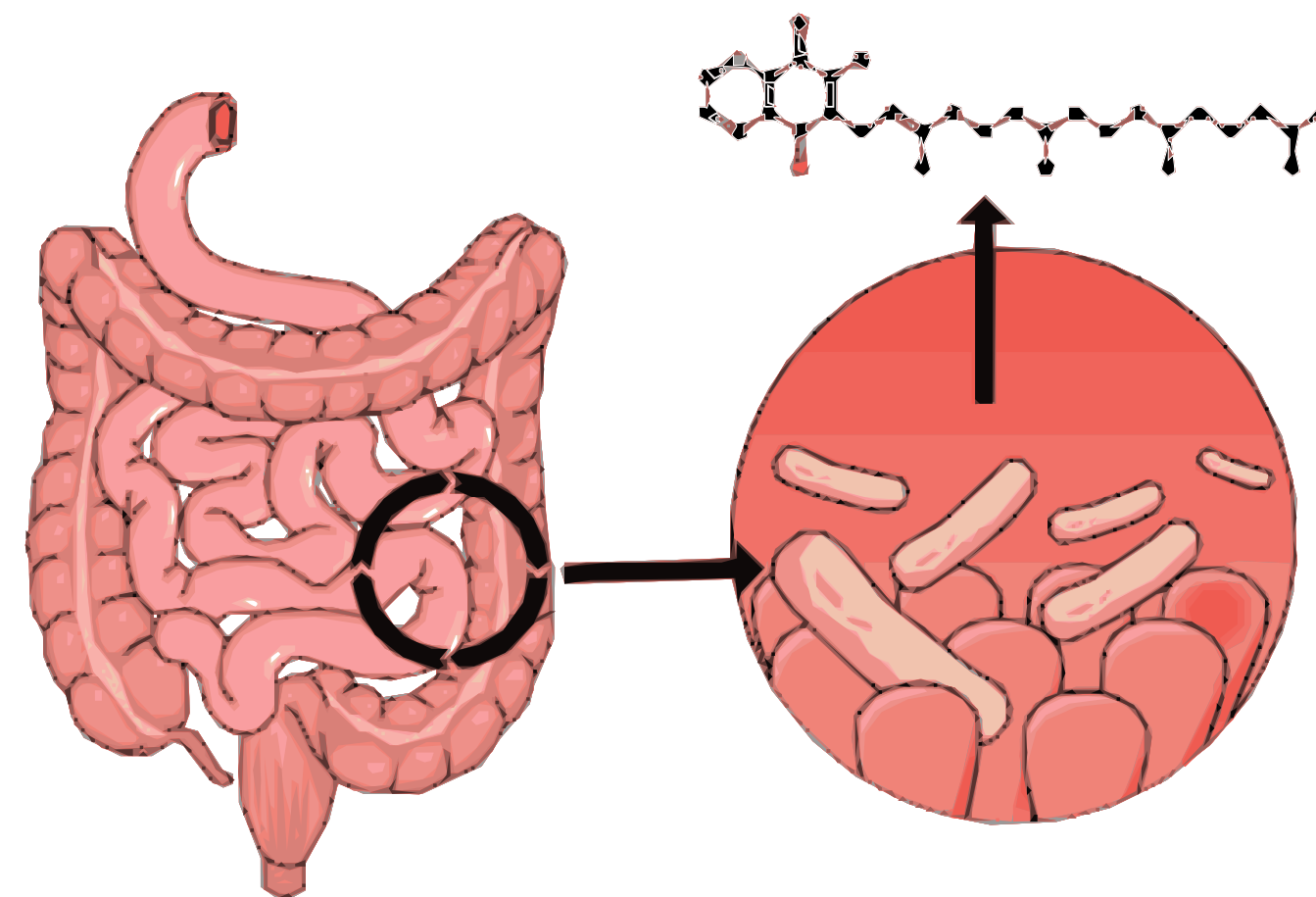
biogeochemical cycles

e.g. N_2



host health/functioning

e.g. vitamin K2
synthesis in gut



*Gilbert JA, Neufeld JD

Life in a World without Microbes. PLoS Biol (2014)

Main question

**what does an organism/cell has to do to be
alive?**

us: quantitative physiology

Cell physiology

is the biological study of the activities that take place in a cell to keep it alive (and their dependency on the environment)

Quantitative cell physiology

is the biological study of the activities that take place in a cell to keep it alive (and their dependency on the environment) **with numbers and equations**

what I want to convince you of

Microbes (biological systems) do interesting and extremely complex things

There exist quantitative biological laws

It is possible to build quantitative, predictive theory in biology

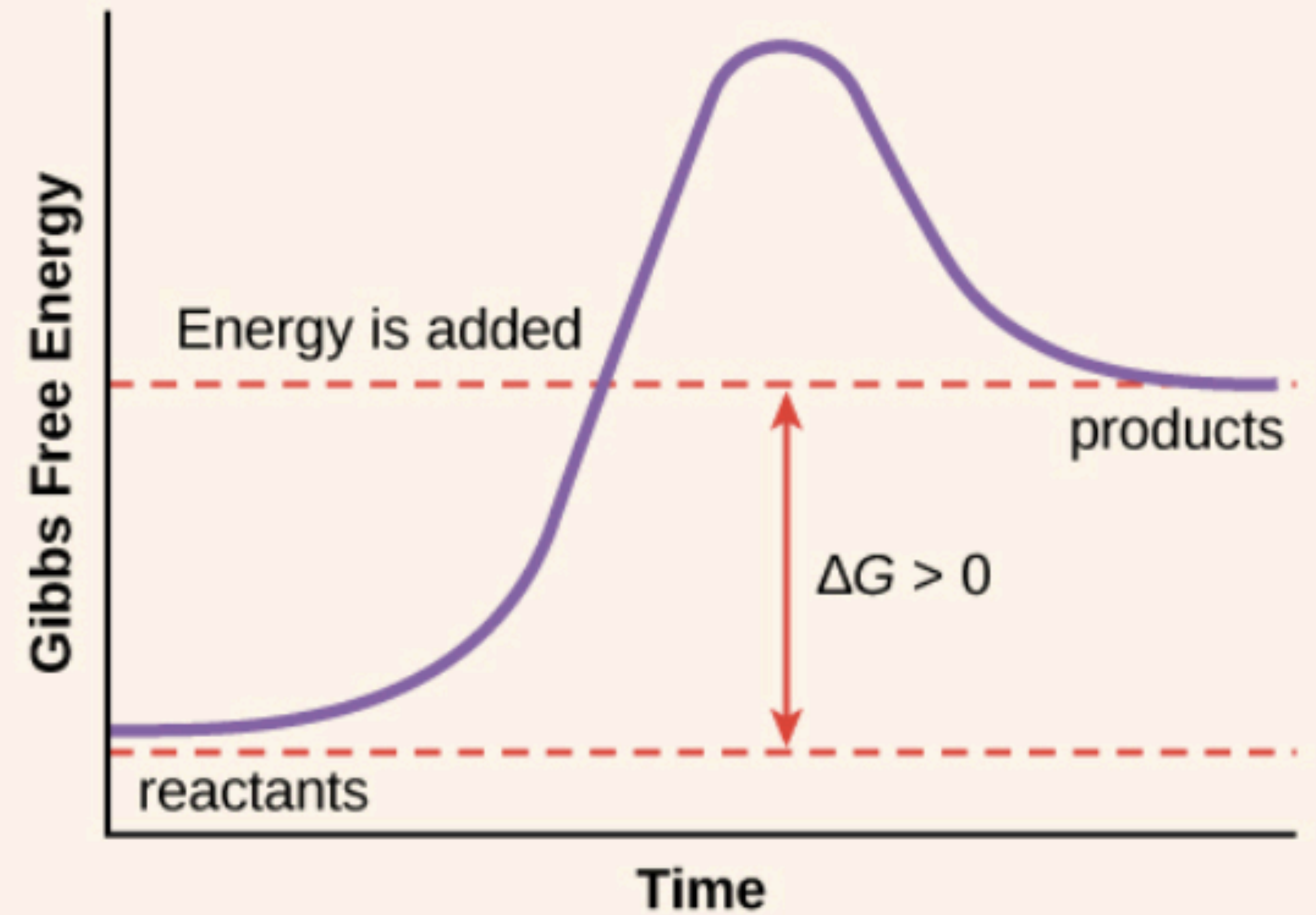
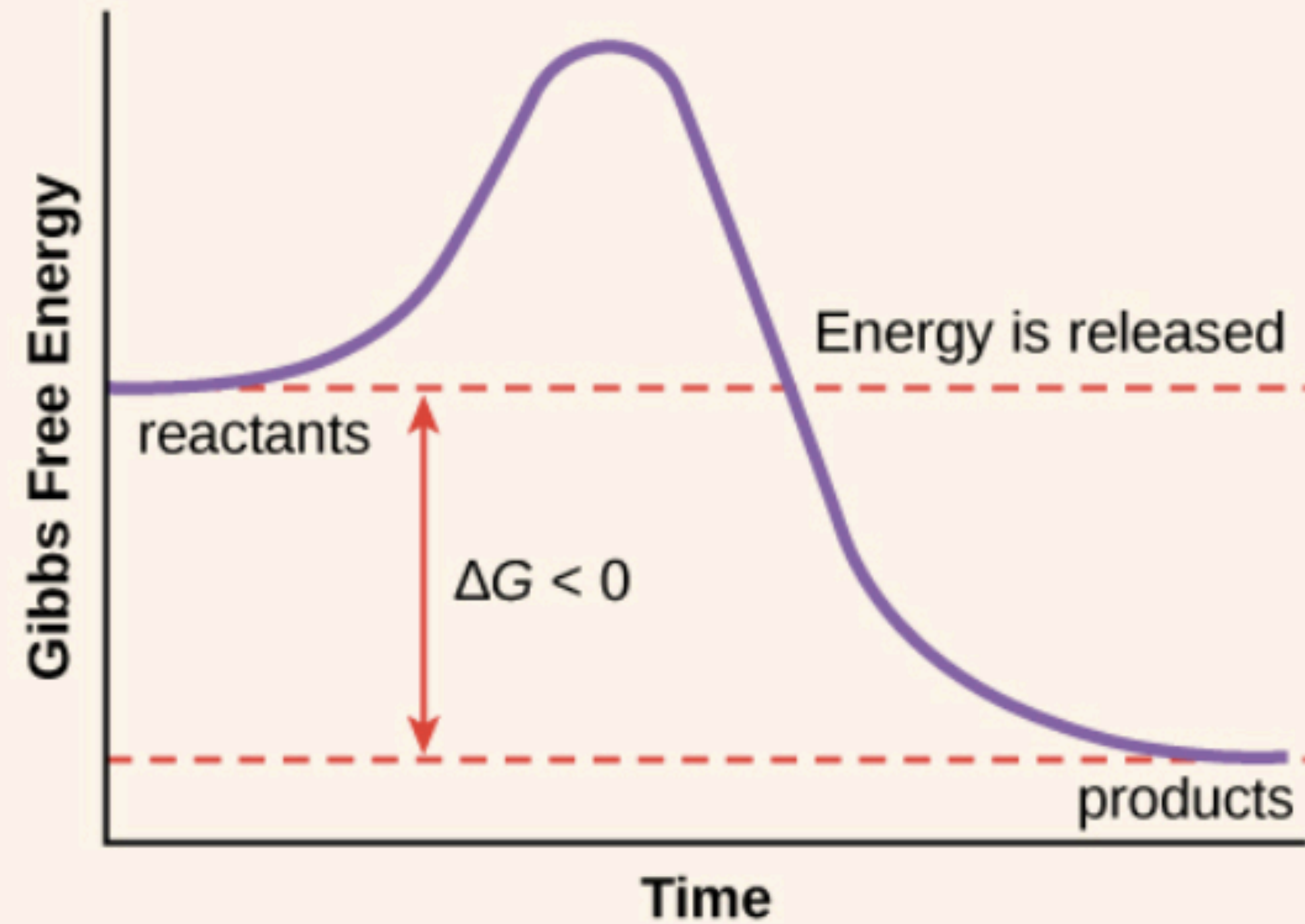
to start: microbial physiology in a nutshell



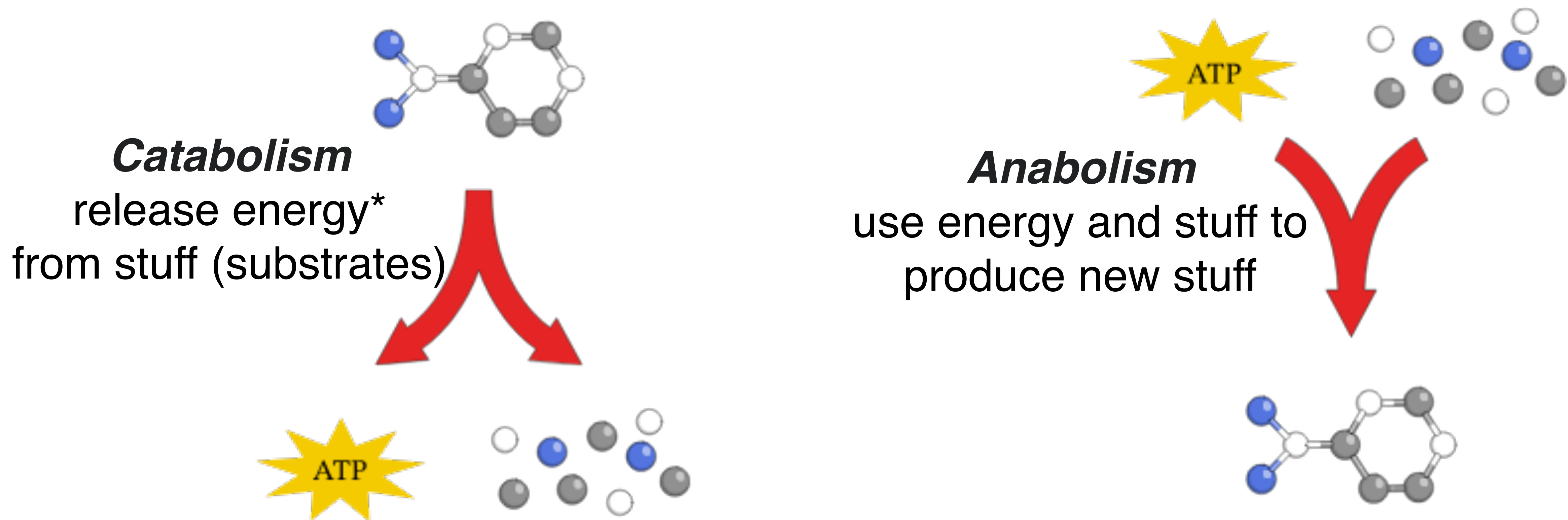
In 5 slides (+ some equations)

150 pages in Brock Biology of Microorganisms (with no equations)

0: There exist chemical reactions

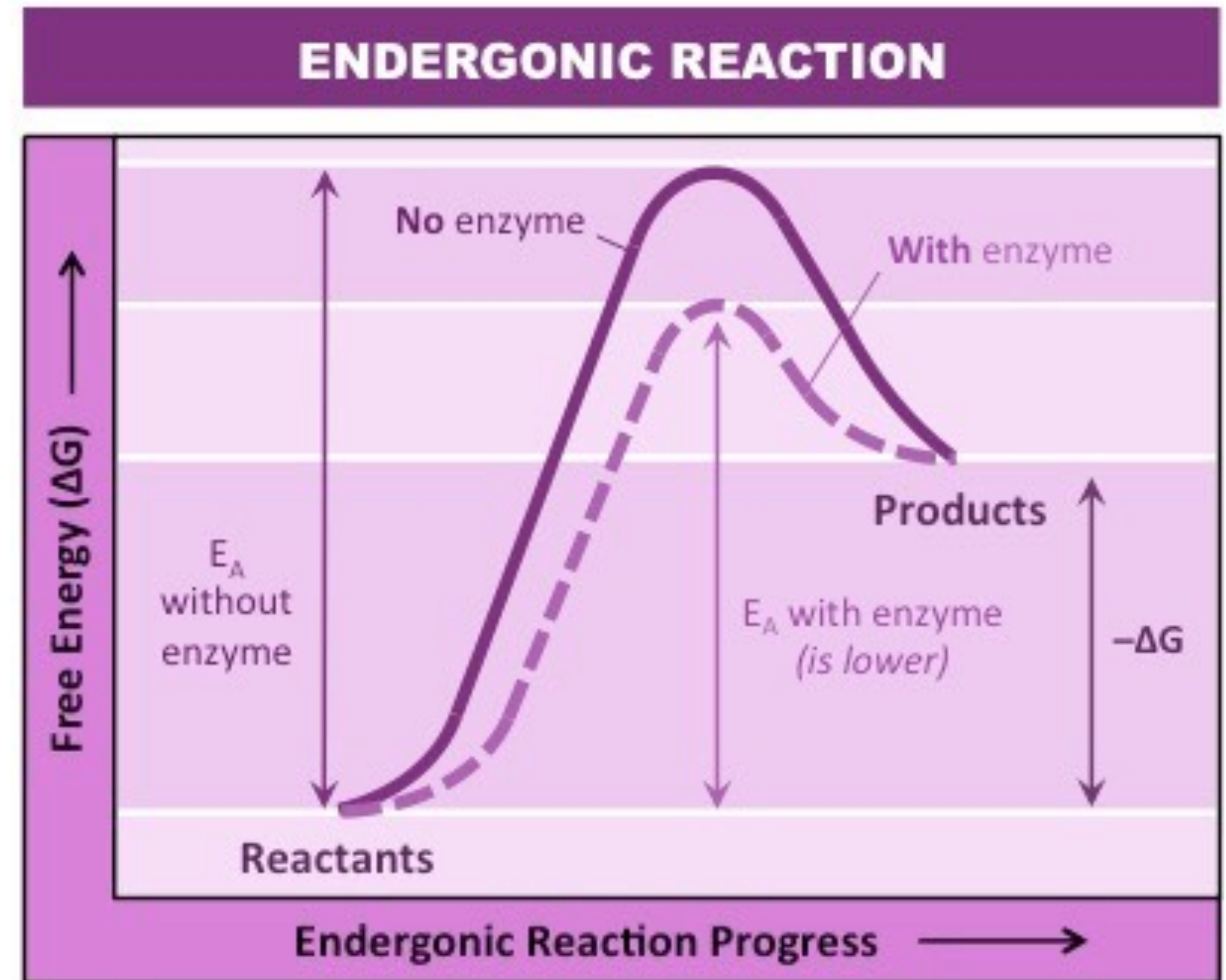
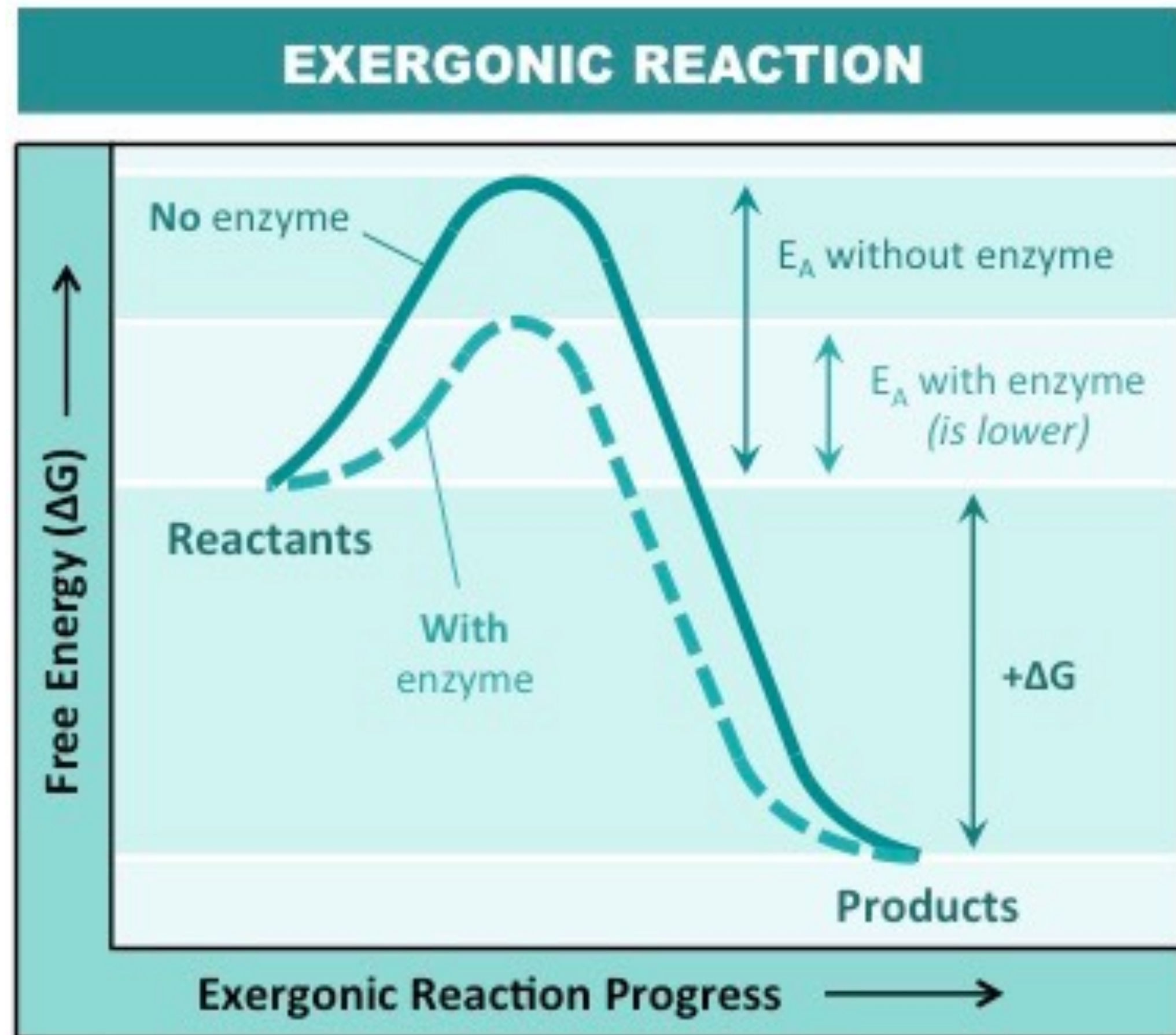


1: Cells perform reactions to get energy and produce new material



**energy can be produced also from light*

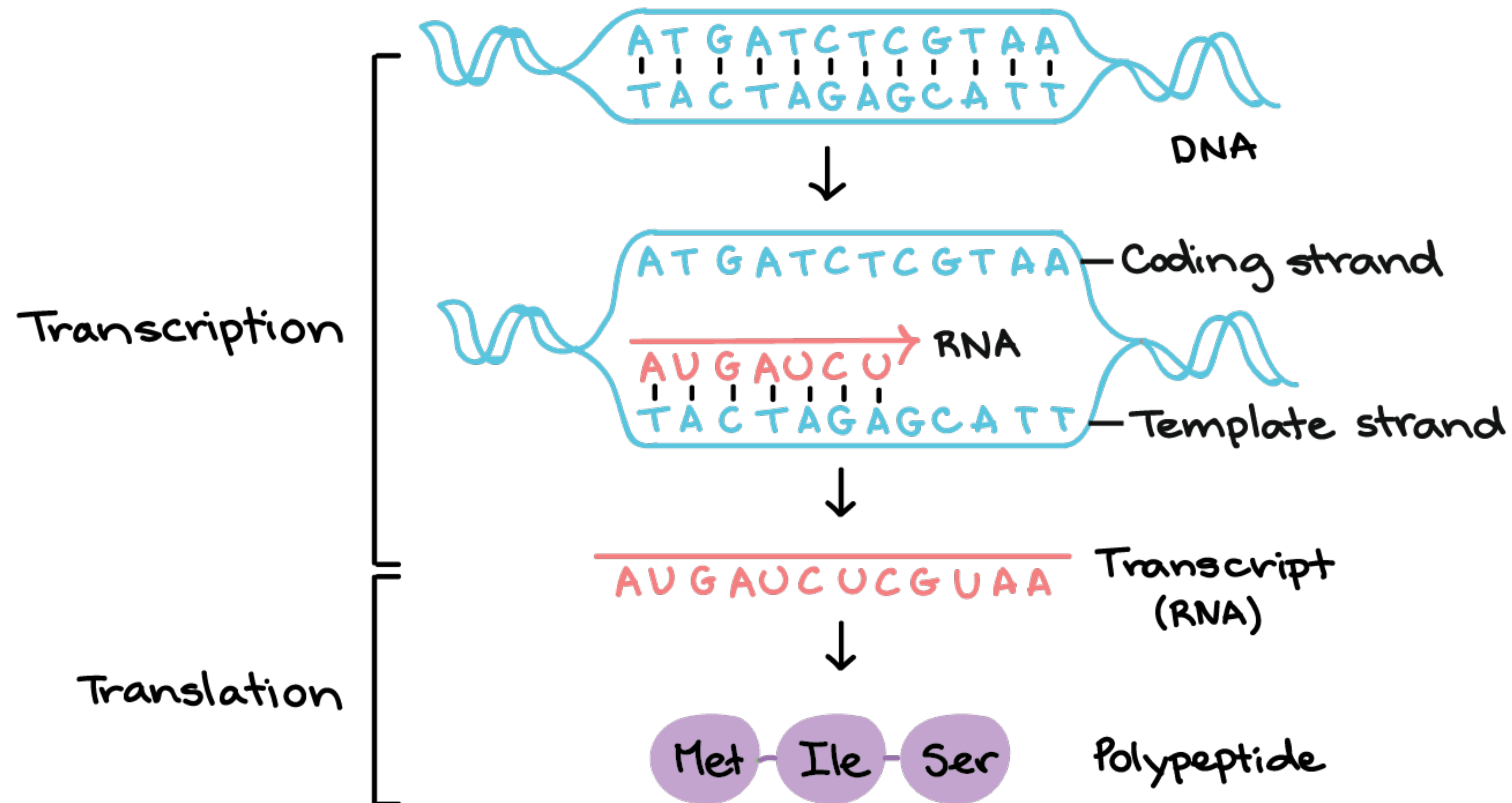
2: Enzymes exist and are specific



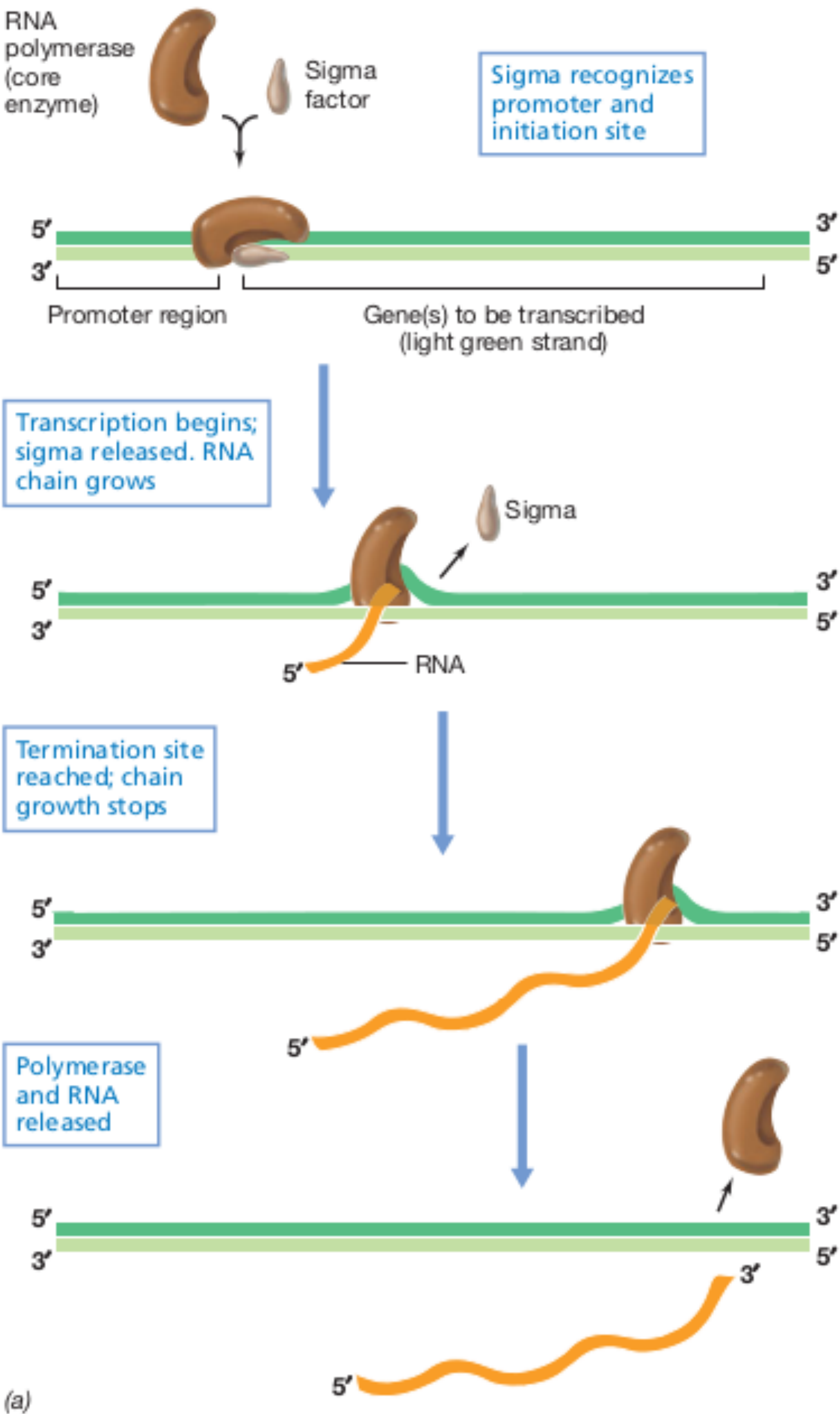
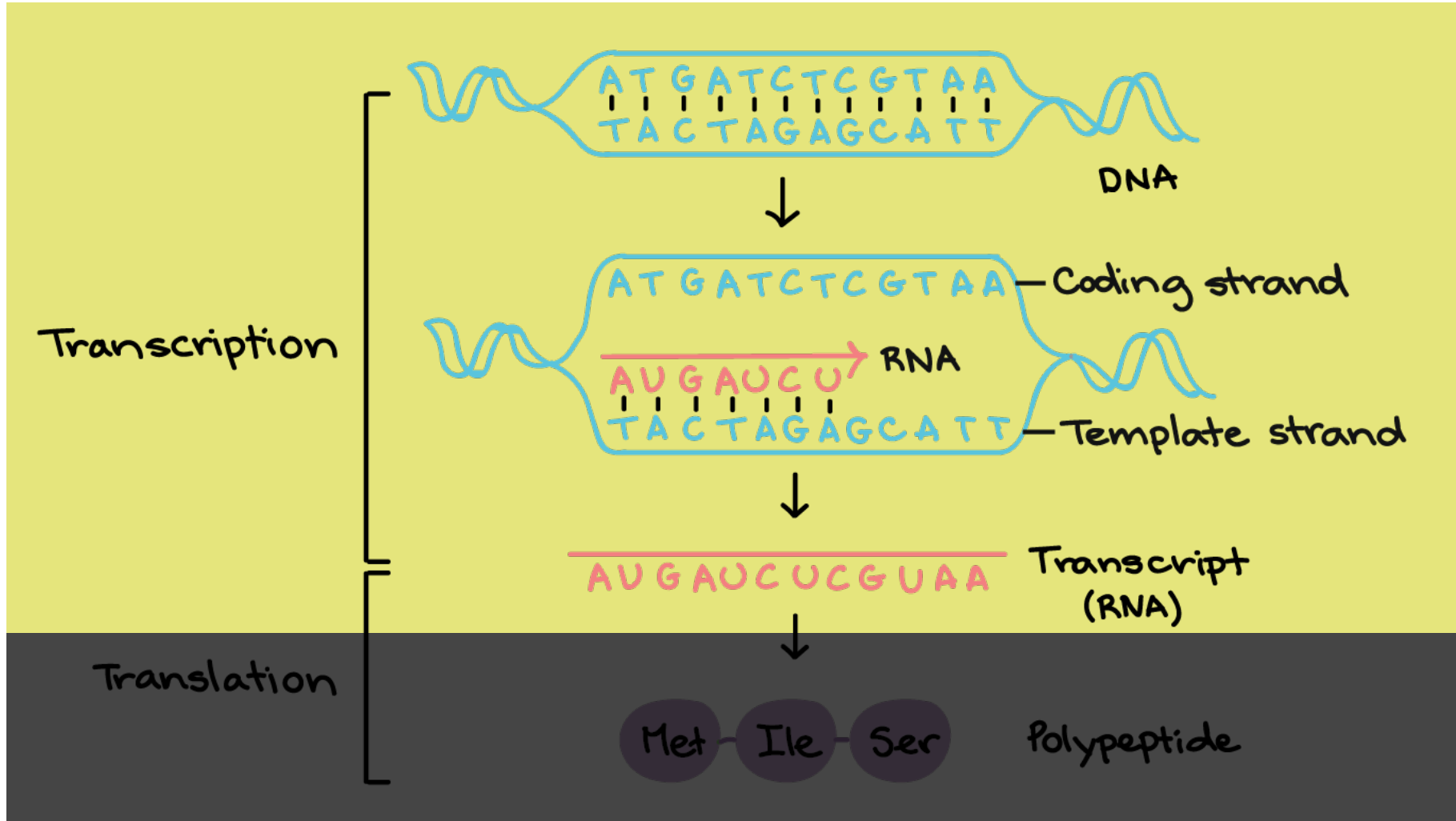
3: The instructions to build enzymes (and every other protein) are written in DNA



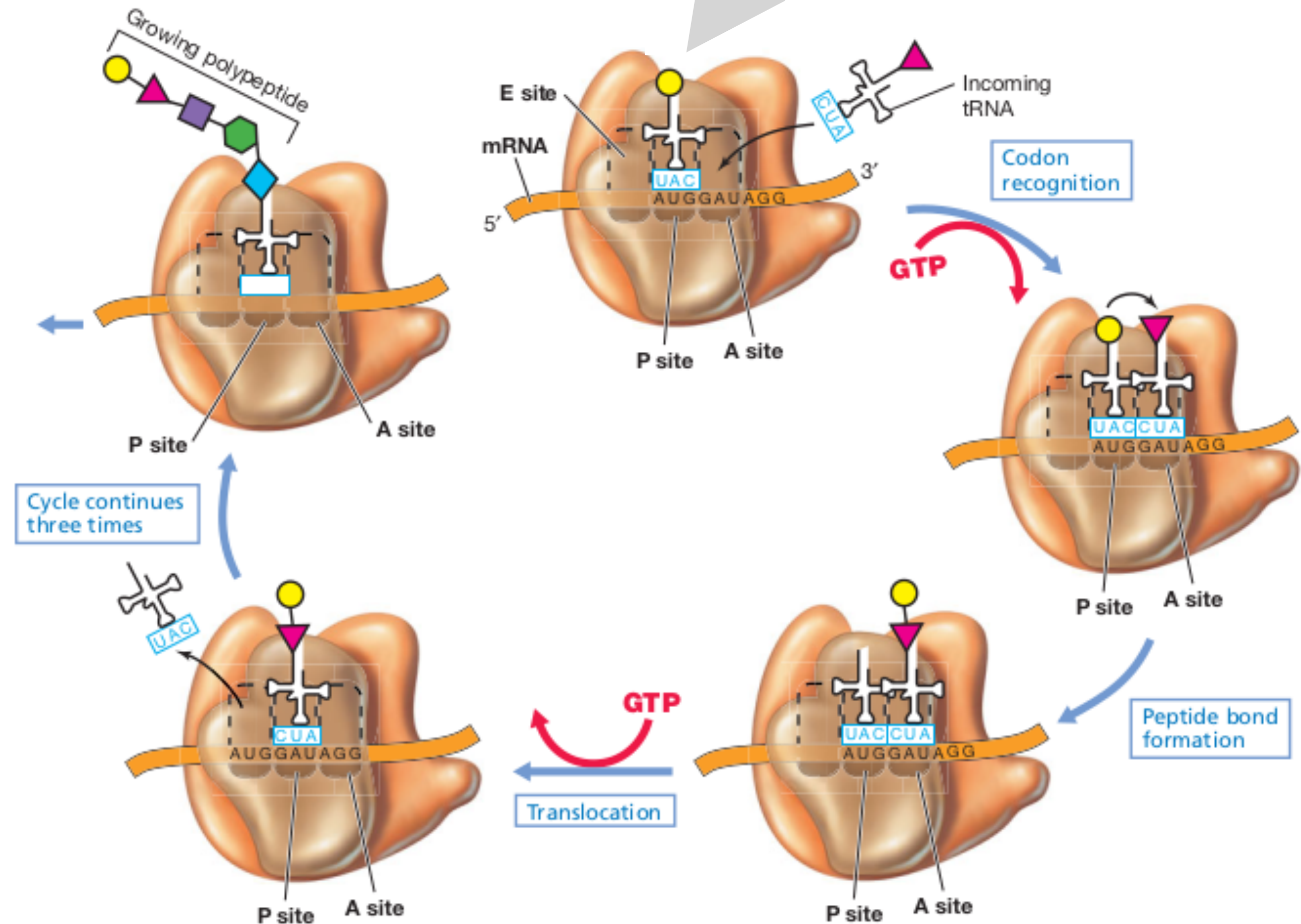
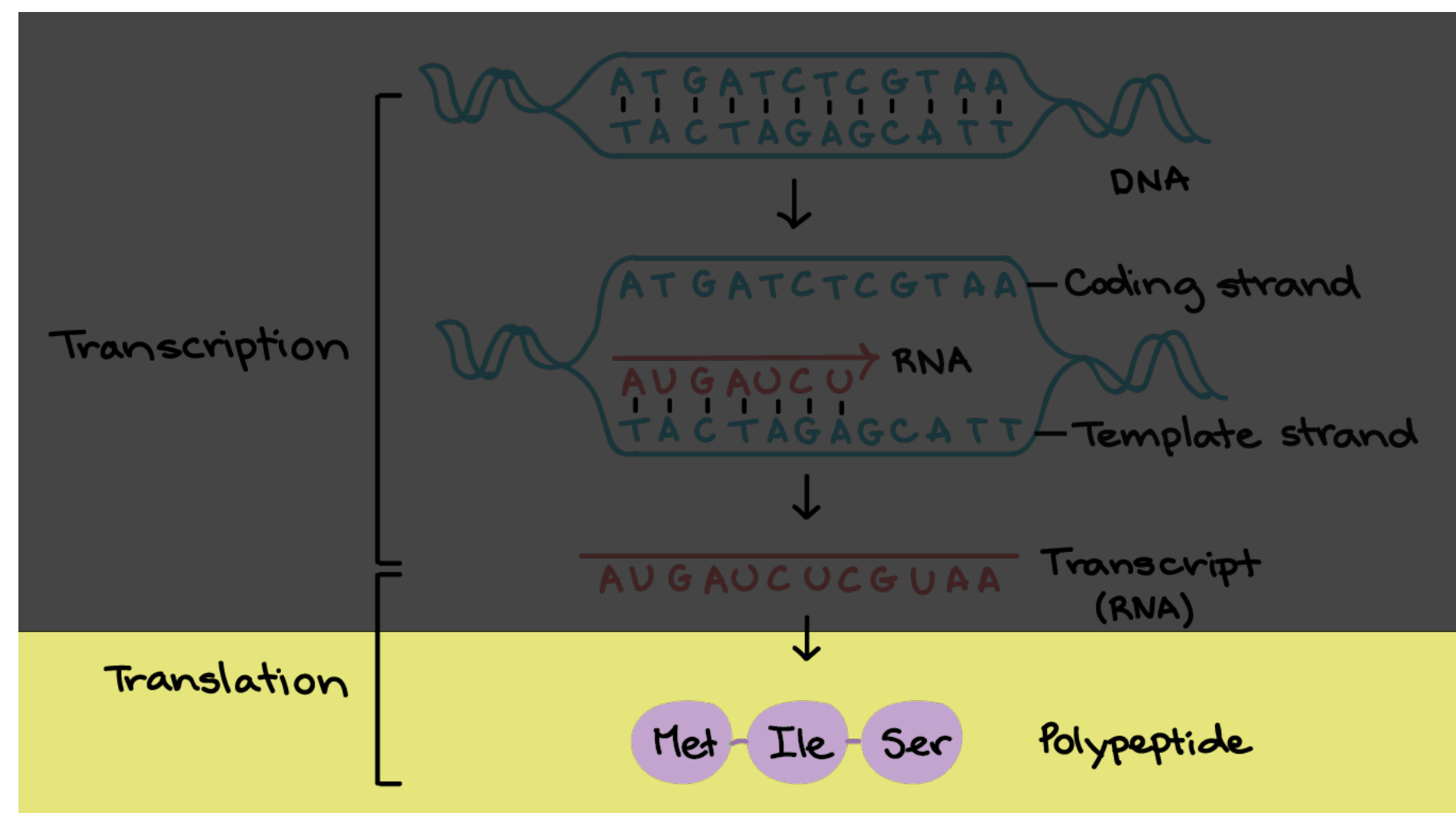
Proteins are expressed in a two step process



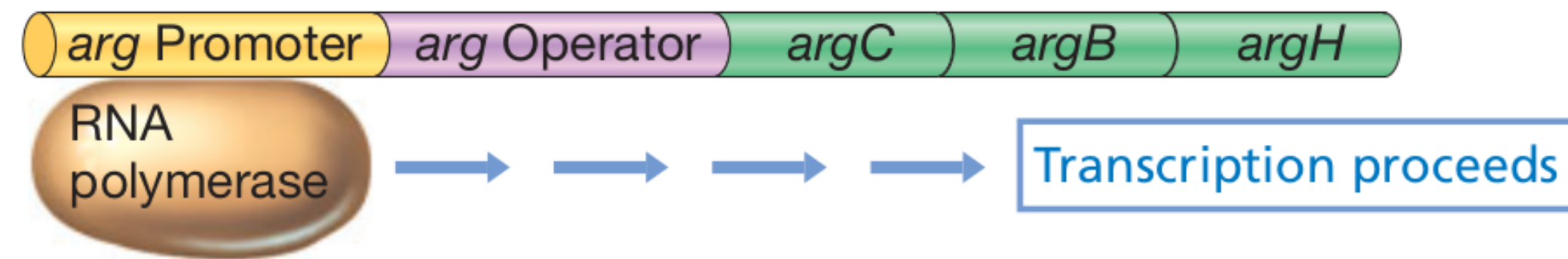
3.1 Transcription from DNA to mRNA



3.2 Translation: ribosomes convert mRNA info to a sequence of aa



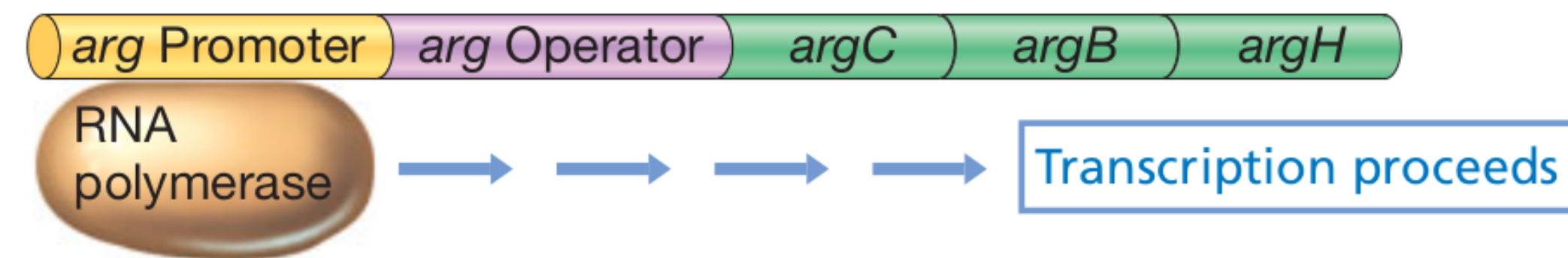
4: Regulation: cells regulate which protein express and how much



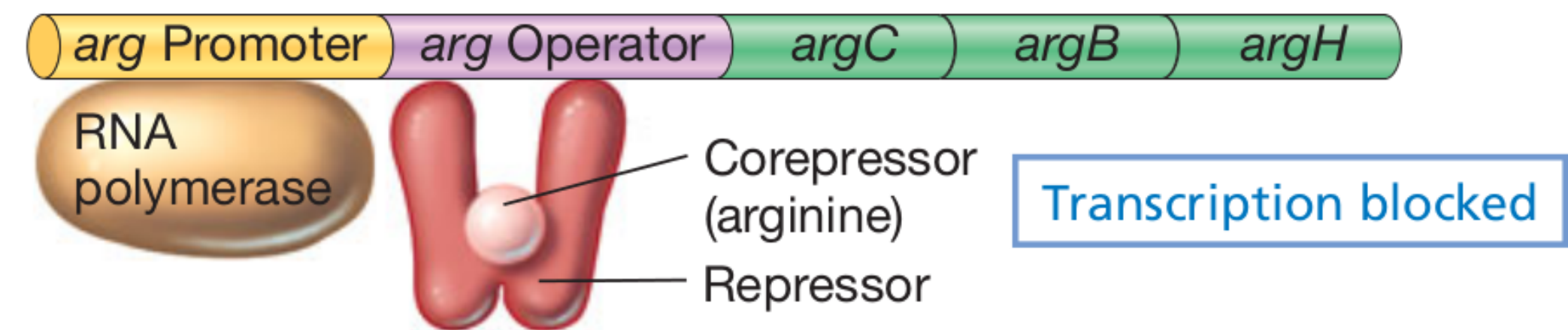
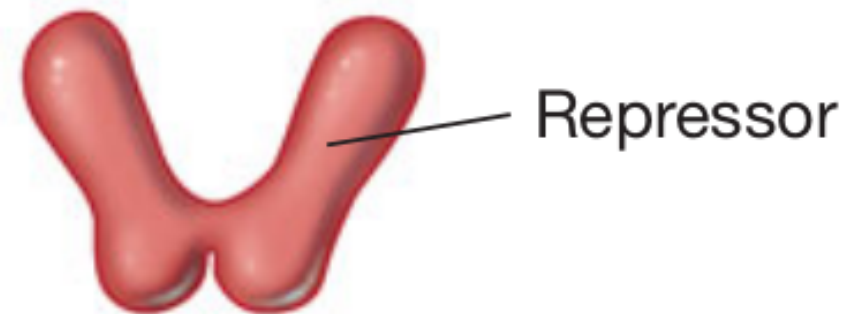
4: Regulation: cells regulate which protein express and how much



Repression (e.g., arginine operon)



(a)

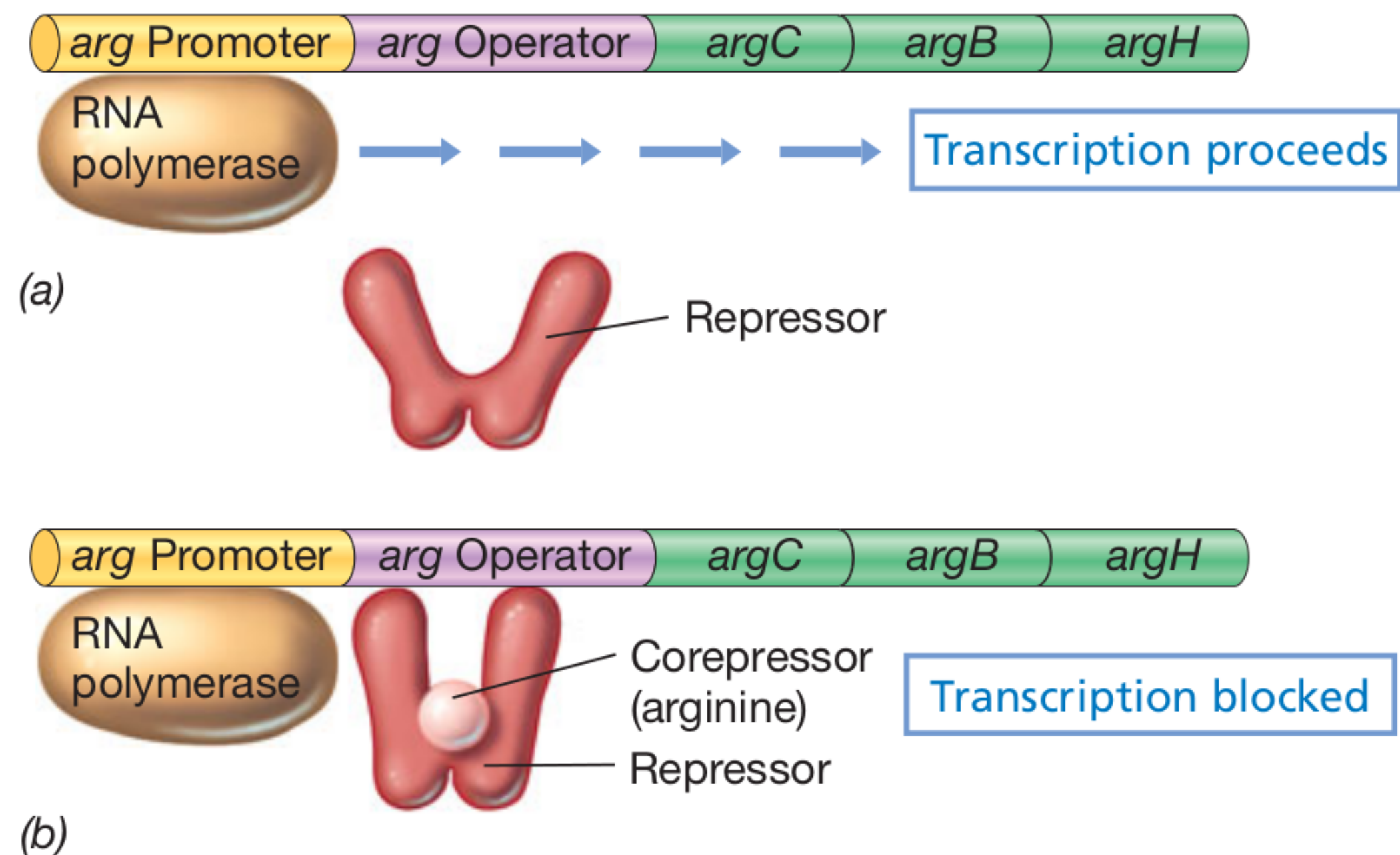


(b)

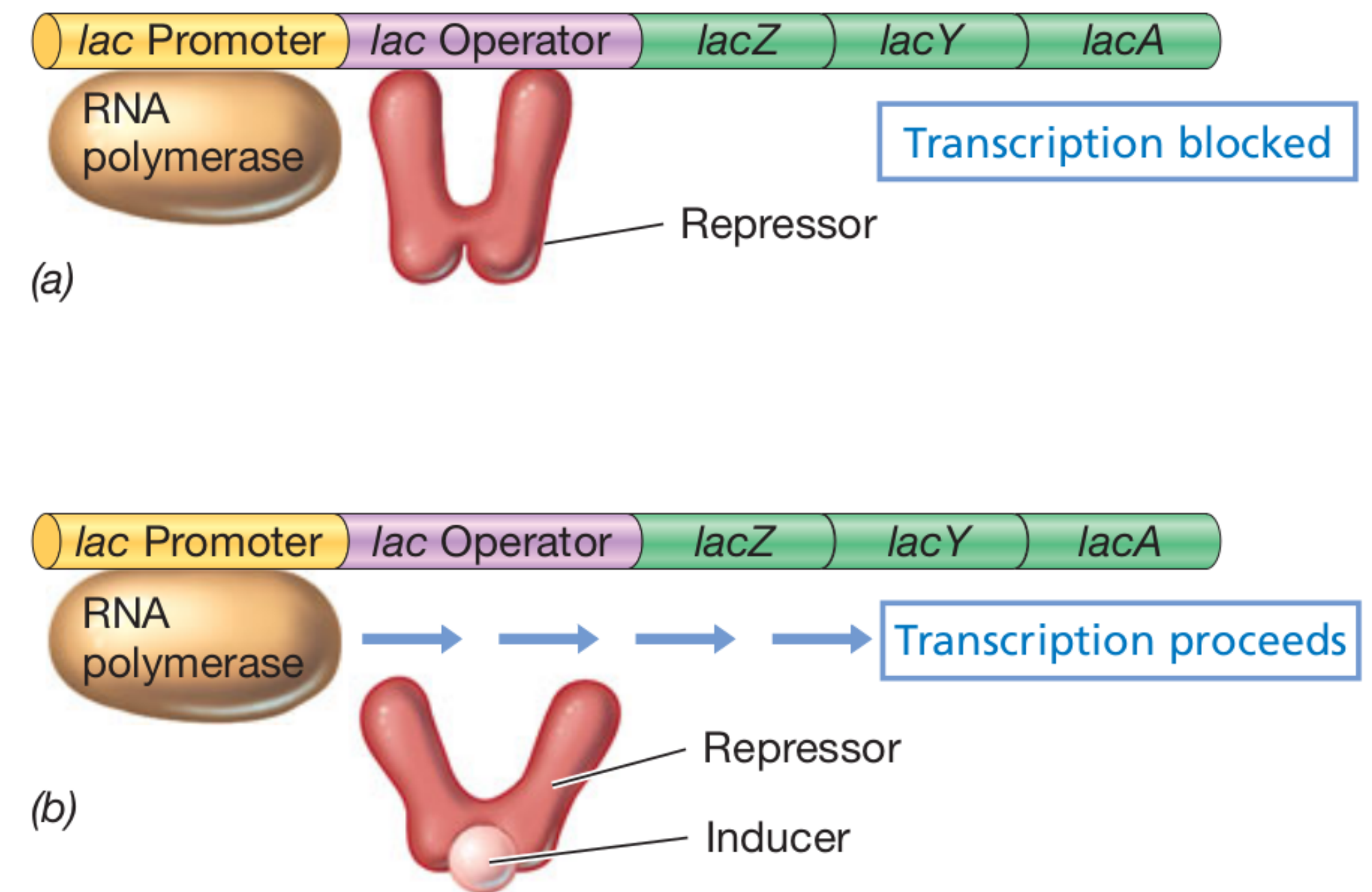
4: Regulation: cells regulate which protein express and how much



Repression (e.g., arginine operon)



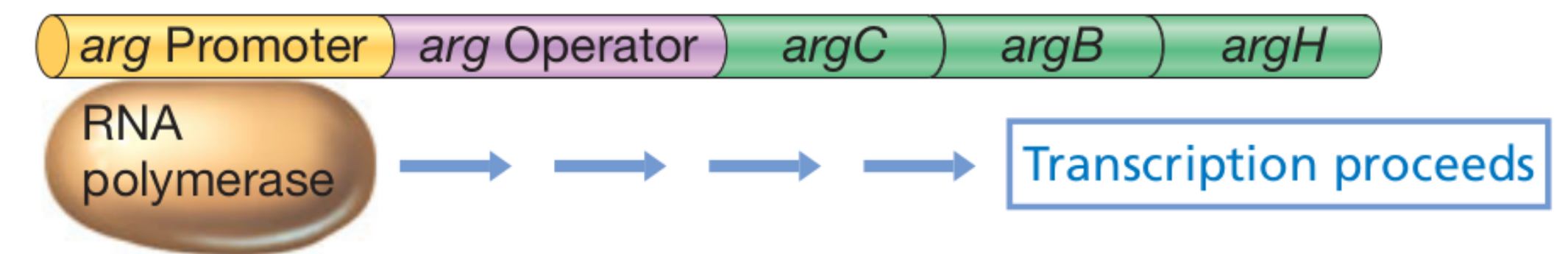
Induction (e.g., lac operon)



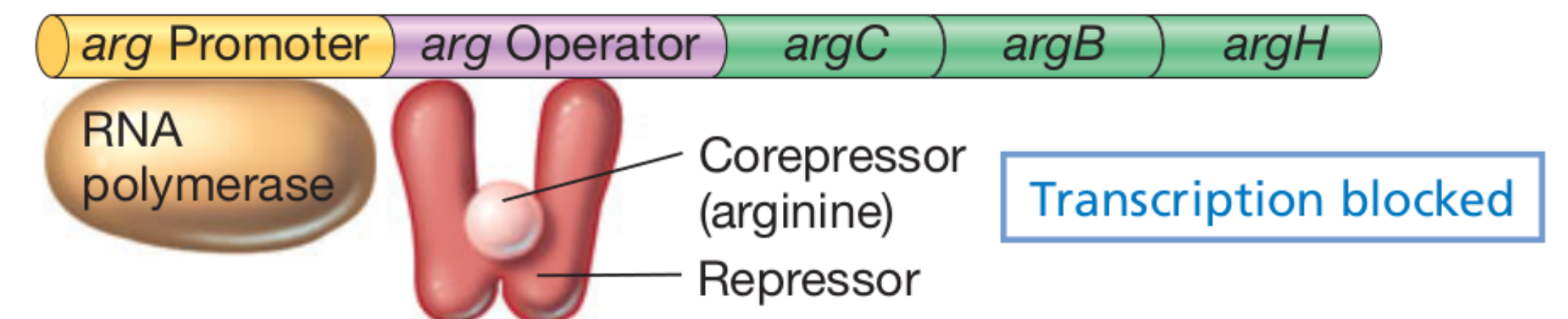
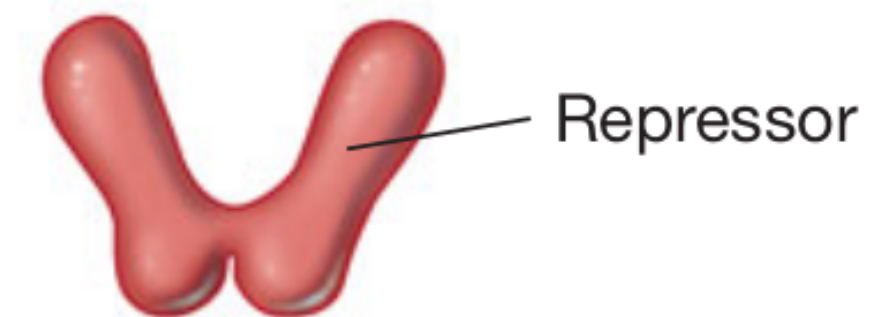
4: Regulation is the way cells take decisions (the example of the arg operon)



The problem from a cell perspective:
Producing aminoacid arginine is costly
when should I produce it?



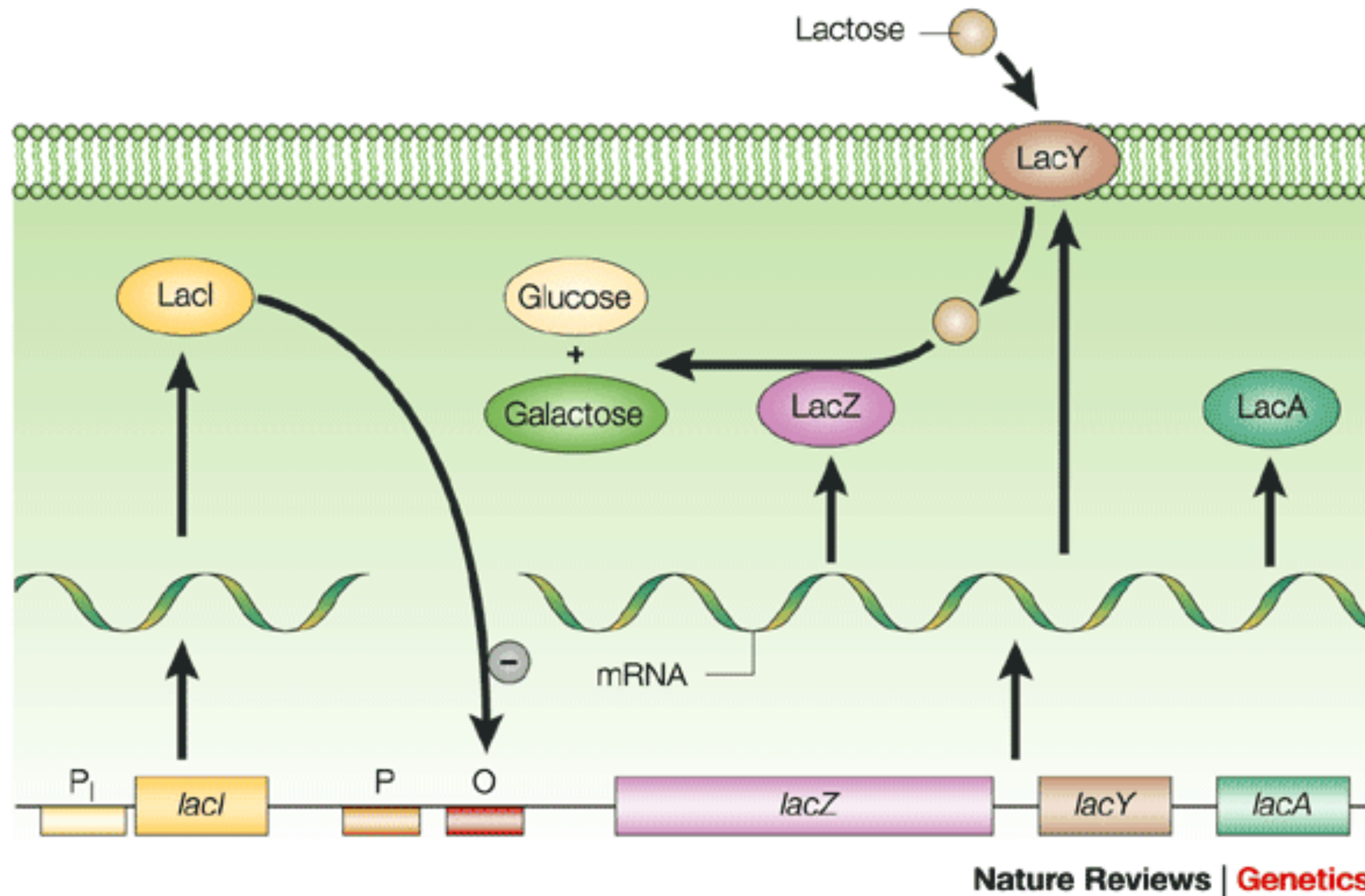
(a)



(b)

4: Regulation is the way cells take decisions

(the example of the lac operon)

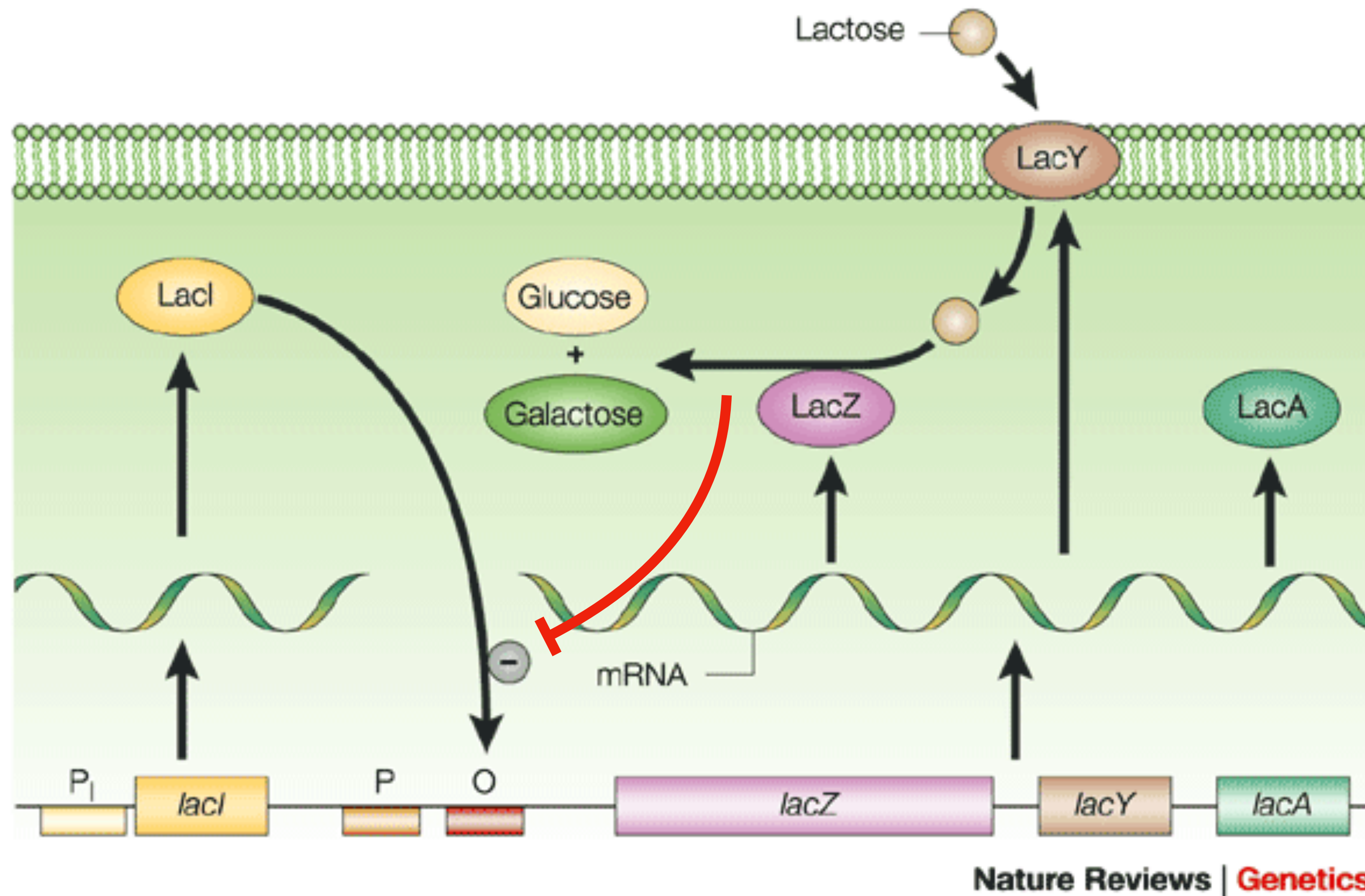


*Monod and Jacob
Nobel prize in 1964*



4: Regulation is the way cells take decisions

(the example of the lac operon)

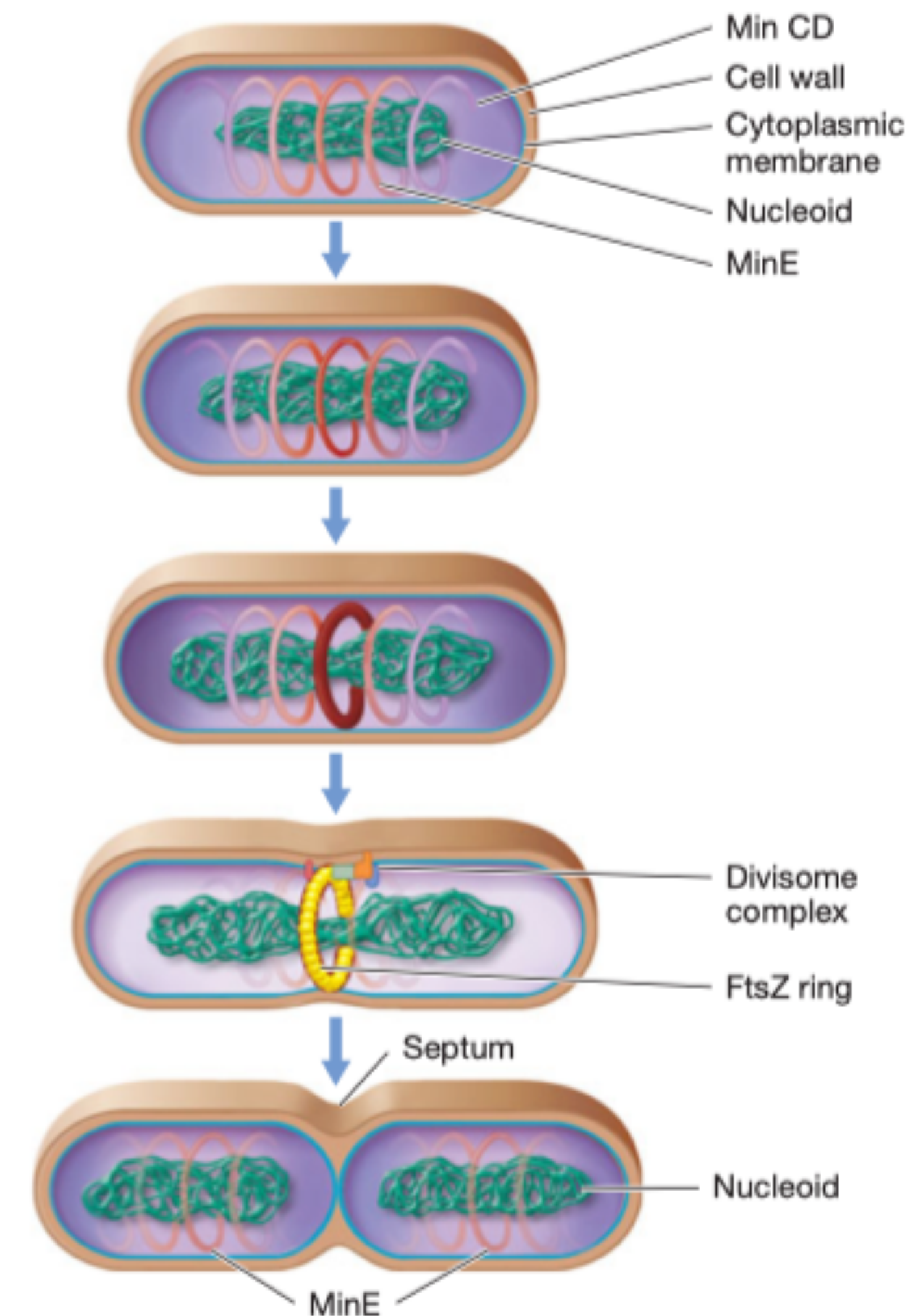
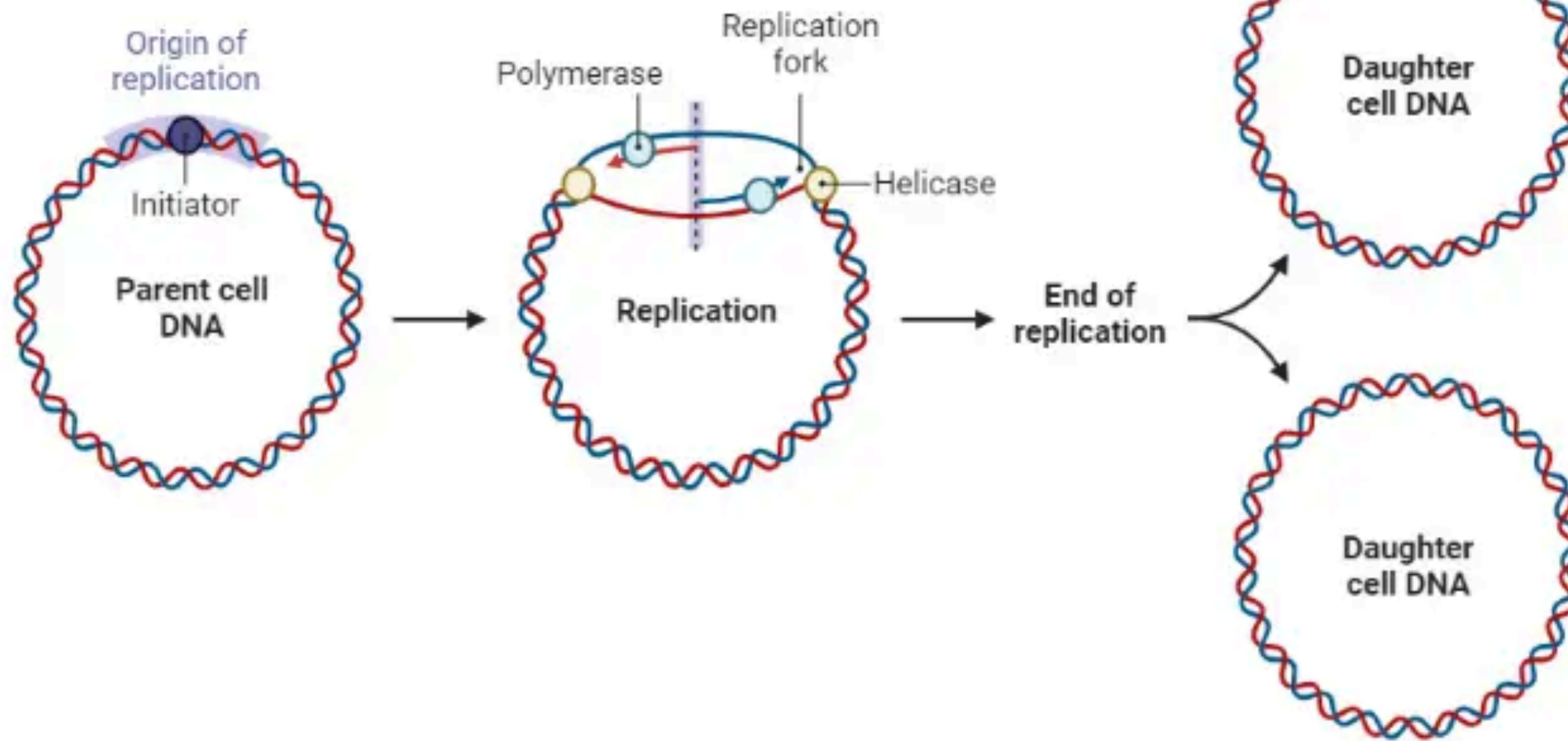


*Monod and Jacob
Nobel prize in 1964*



5: DNA is replicated and a cell divides

“The dream of every cell is to become two cells” - F. Jacob



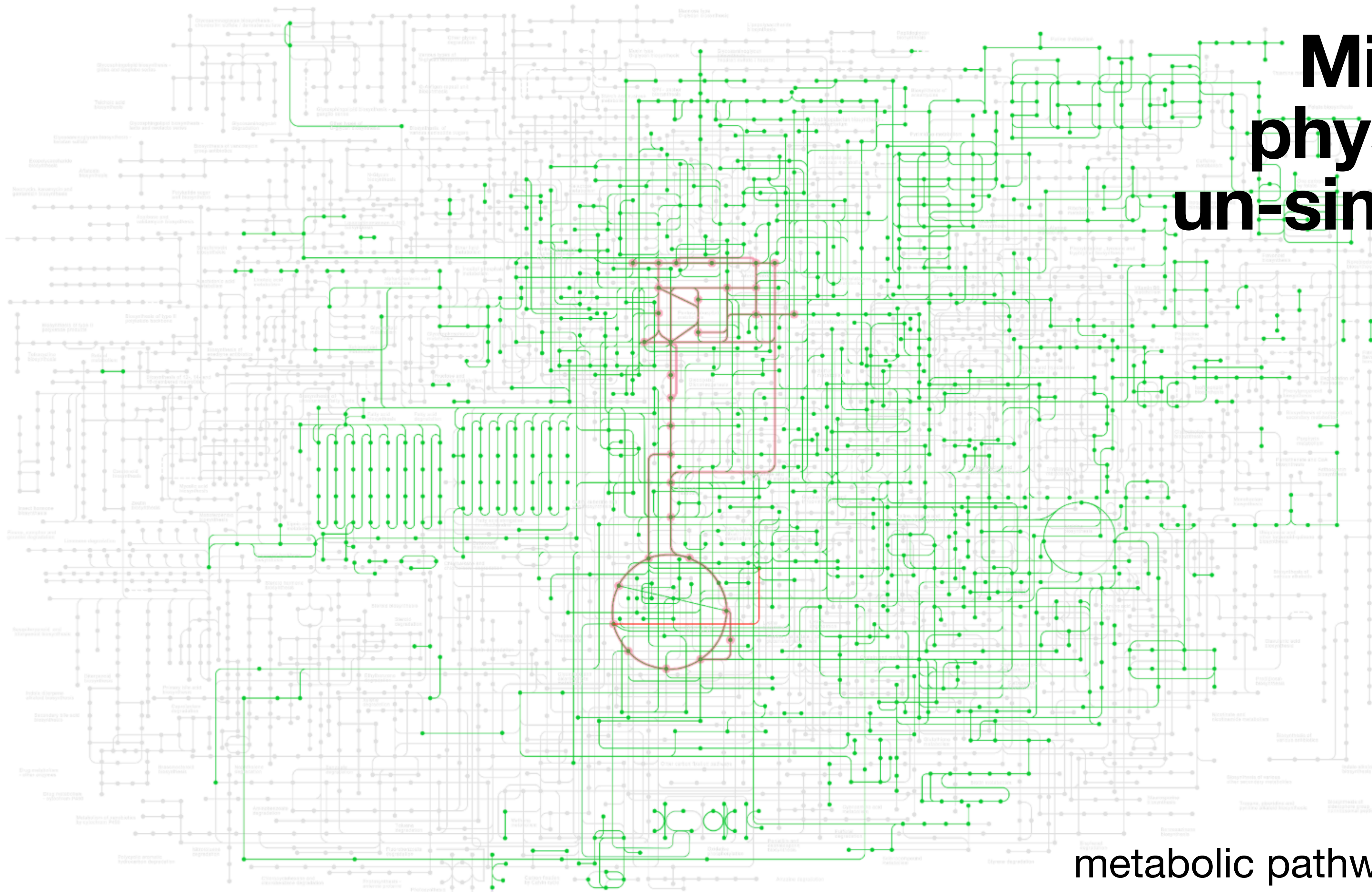
More about this in Ariel Amir's lectures

Microbial physiology in a nutshell



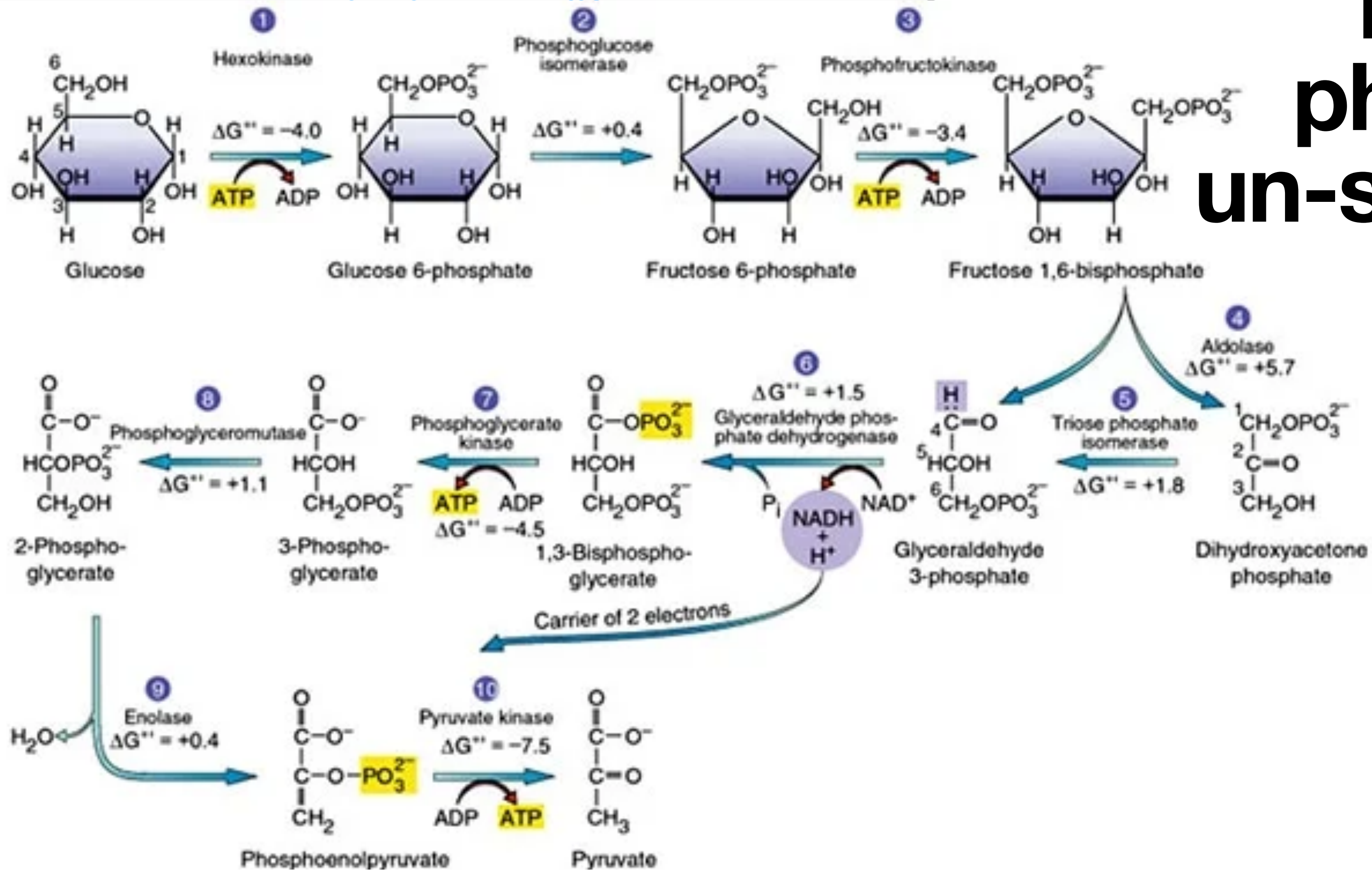
1. Cells perform reactions to get energy and produce new material
2. Enzymes exist and are specific
3. The instructions to build enzymes (and every other protein) are written in DNA
4. Cells take decisions by regulating which proteins express and how much
5. Cells replicate DNA and become two cells

Microbial physiology un-simplified

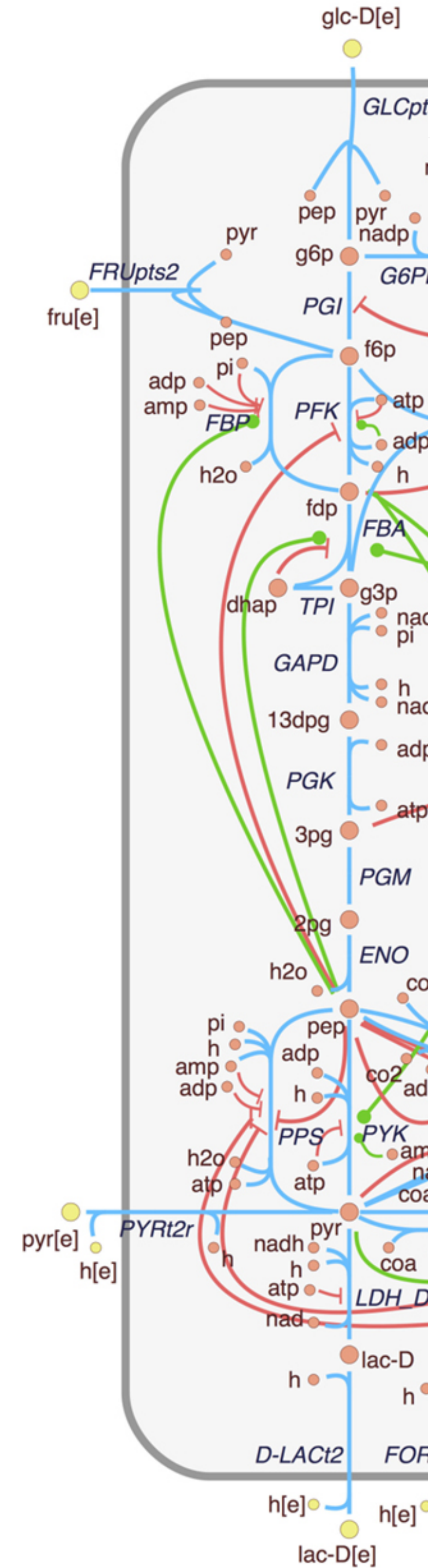


metabolic pathways in *E. coli*

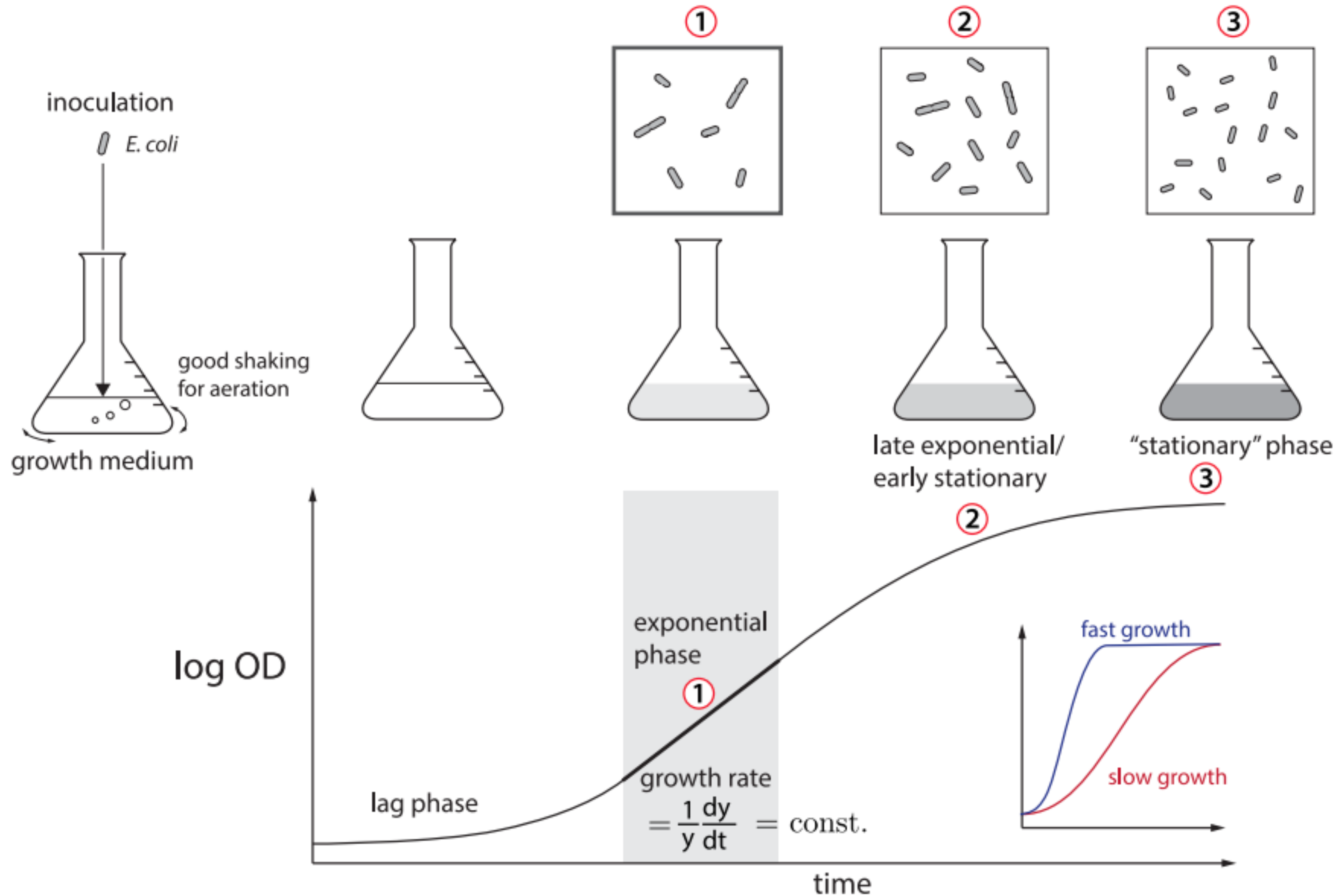
Microbial physiology un-simplified



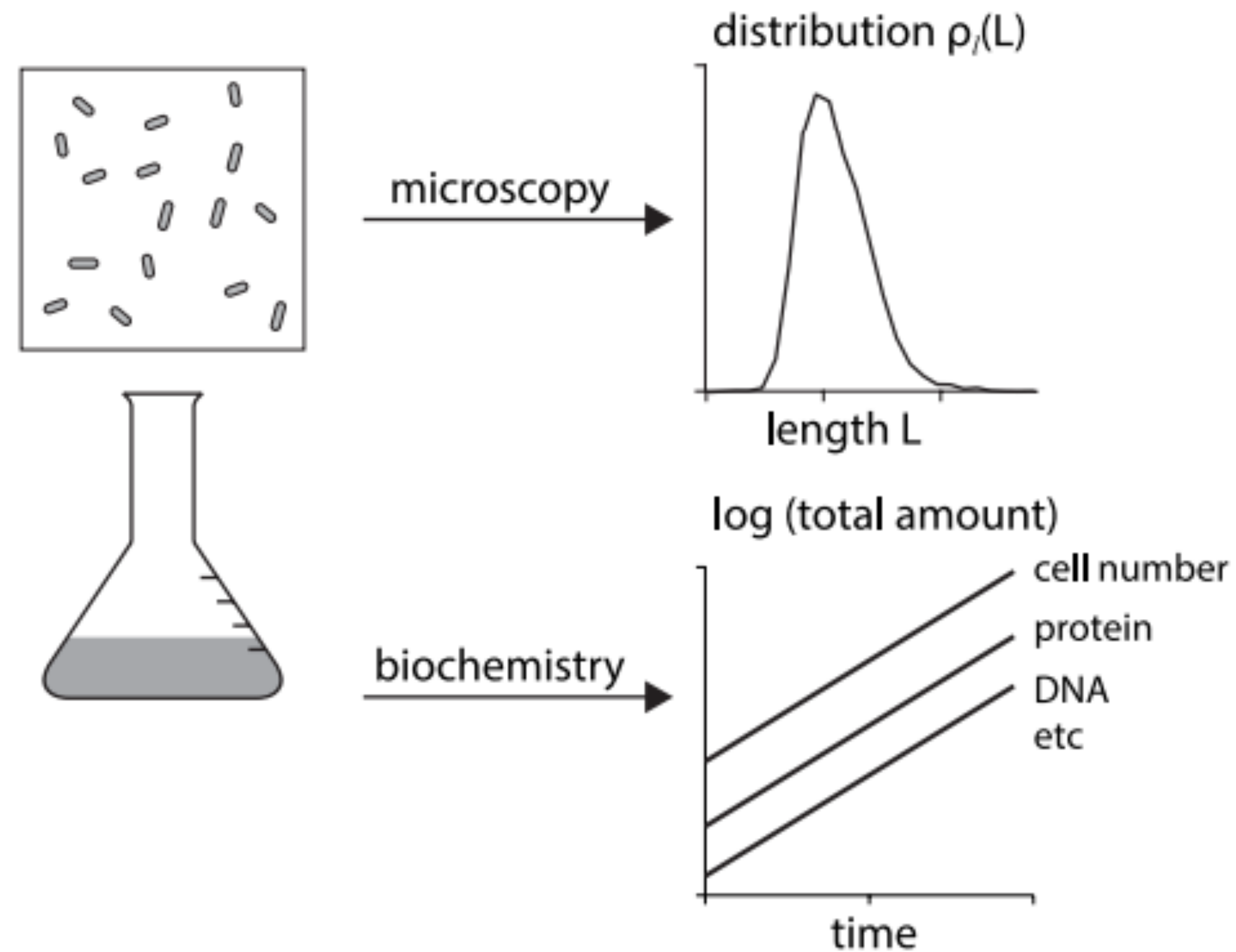
Glycolysis



Growing bacteria in the lab

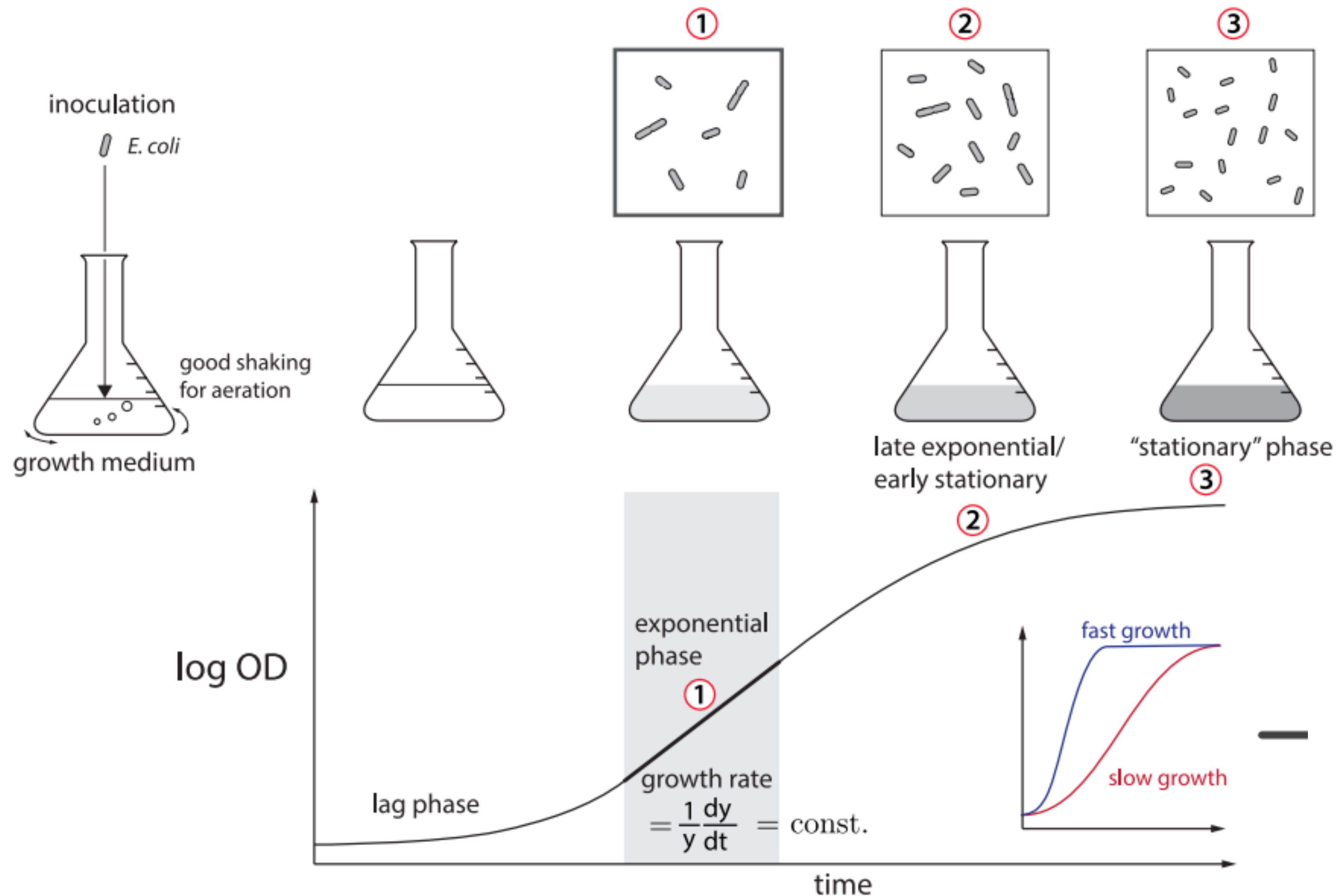


what else can we measure?



when is easier to study physiology?

physiology: biological study of the activities that take place in a cell to keep it alive



References

Jun et al., 2018 Rep. Prog. Phys. 81 056601

Brock Biology of Microorganisms